ABCD ADHD menarche analysis: Output for cross-sectional models

Model 1 output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.702   0.00322 0.696 0.708


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ menarche_status_p + age_sc + ethnrace + inr_sc + (1 | participant_id)
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.830     0.045    0.737    0.914 1.003      665     1190

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.042     0.035    0.974    1.110 1.002      835     2037

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.177     0.040    0.114    0.265 1.000     7021     6366

Regression Coefficients:
                   Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept             0.038     0.051   -0.061    0.136 1.000     7609     6068
menarche_status_pY    0.023     0.018   -0.013    0.058 1.000    16051     5994
```

```
age_sc               -0.040     0.016   -0.070   -0.009 1.000    14345     6338
ethnraceblack_nh      0.019     0.064   -0.108    0.145 1.000    11360     5496
ethnracehispanic     -0.048     0.060   -0.168    0.068 1.000    10992     6237
ethnraceother         0.055     0.069   -0.079    0.187 1.001    12117     6097
inr_sc               -0.084     0.023   -0.129   -0.041 1.001    18061     5754

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

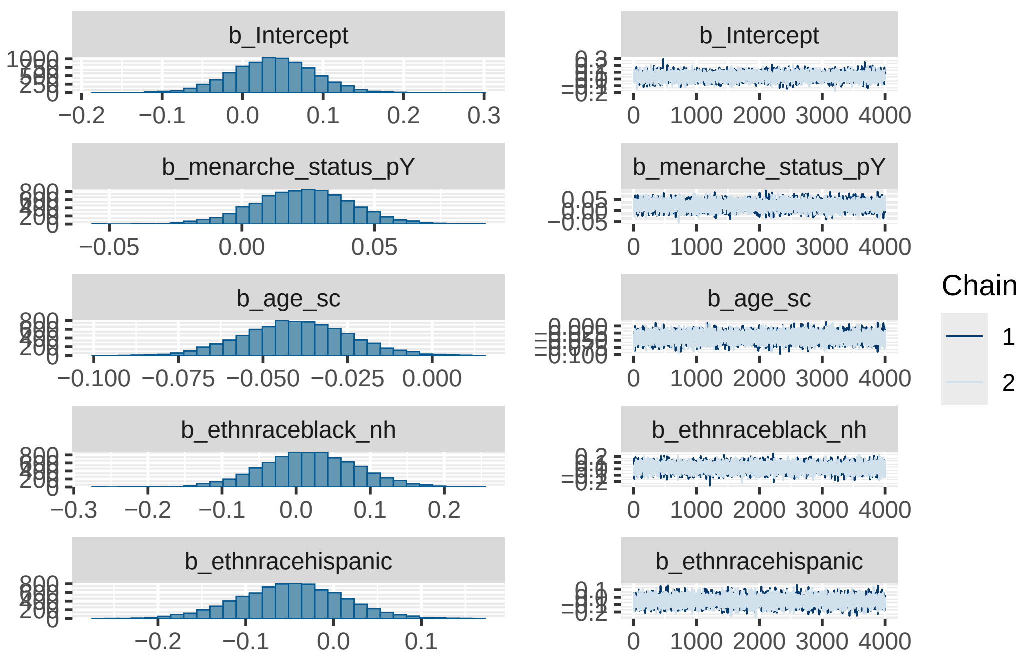
Model 1 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

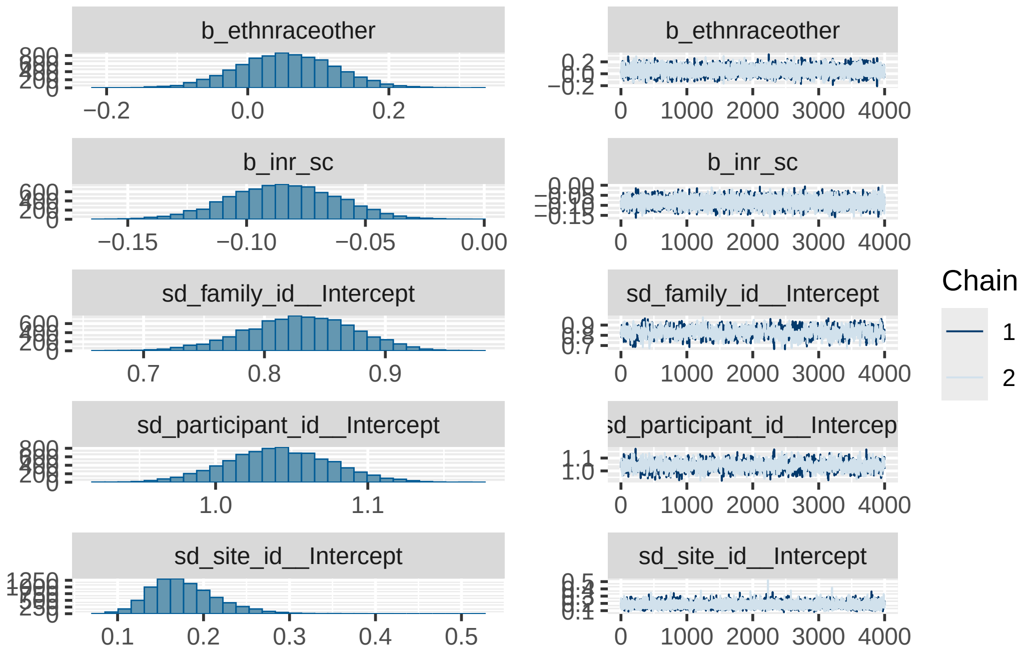
```
# A tibble: 1 x 4
  Variable   Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>         <dbl>          <dbl>          <dbl>
1 Menarche Y     2.33          -1.29           5.99
```

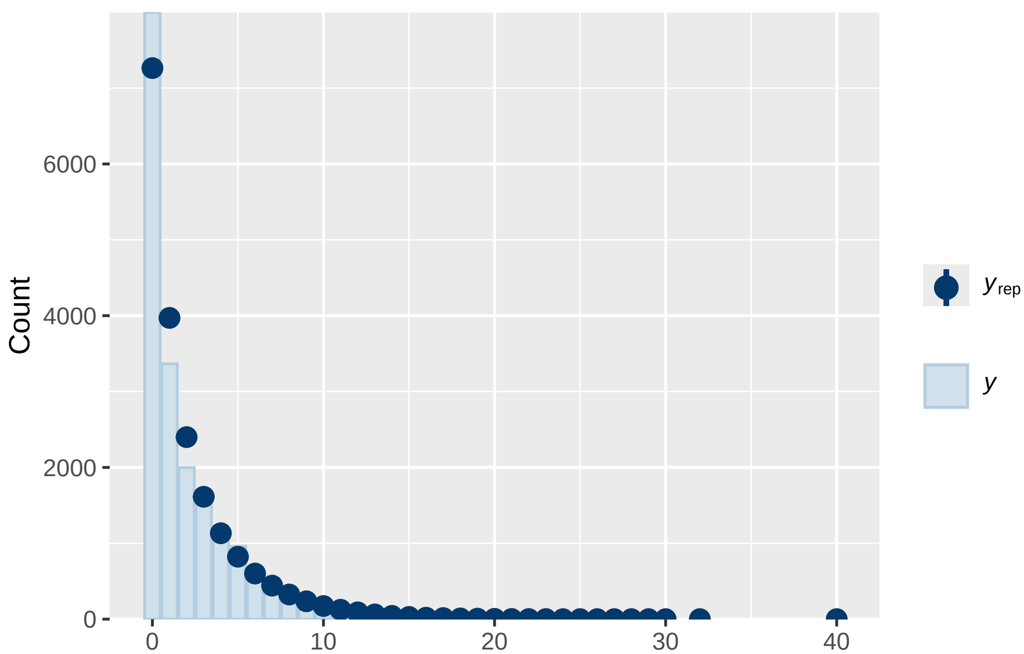
Model 1 diagnostics

```
                prior     class               coef          group resp dpar
         normal(0, 1)         b
         normal(0, 1)         b             age_sc
         normal(0, 1)         b   ethnraceblack_nh
         normal(0, 1)         b   ethnracehispanic
         normal(0, 1)         b      ethnraceother
         normal(0, 1)         b             inr_sc
         normal(0, 1)         b menarche_status_pY
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                         family_id
 student_t(3, 0, 2.7)        sd          Intercept      family_id
 student_t(3, 0, 2.7)        sd                    participant_id
 student_t(3, 0, 2.7)        sd          Intercept participant_id
 student_t(3, 0, 2.7)        sd                           site_id
 student_t(3, 0, 2.7)        sd          Intercept        site_id
 nlpar lb ub tag       source
                         user
                 (vectorized)
                 (vectorized)
                 (vectorized)
```

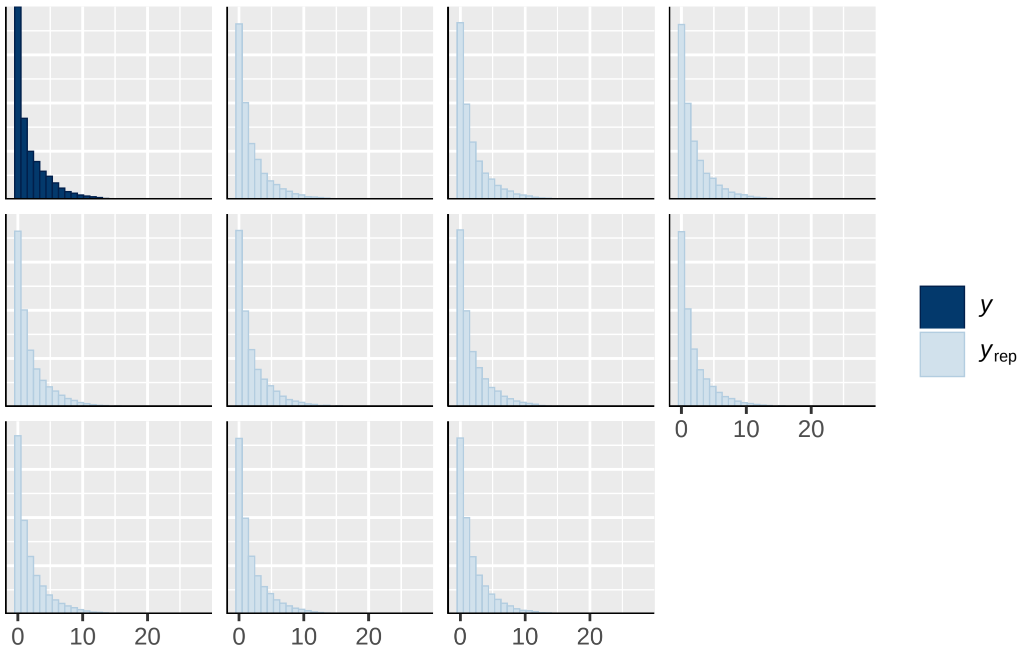
```
                    (vectorized)
                    (vectorized)
                    (vectorized)
                         default
0                        default
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
```



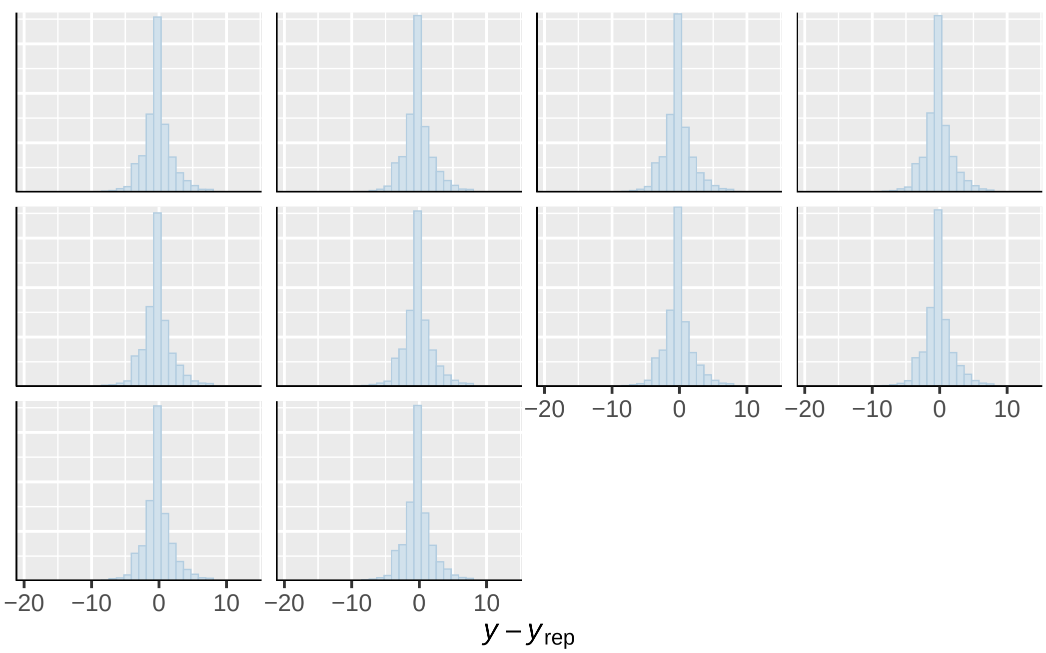




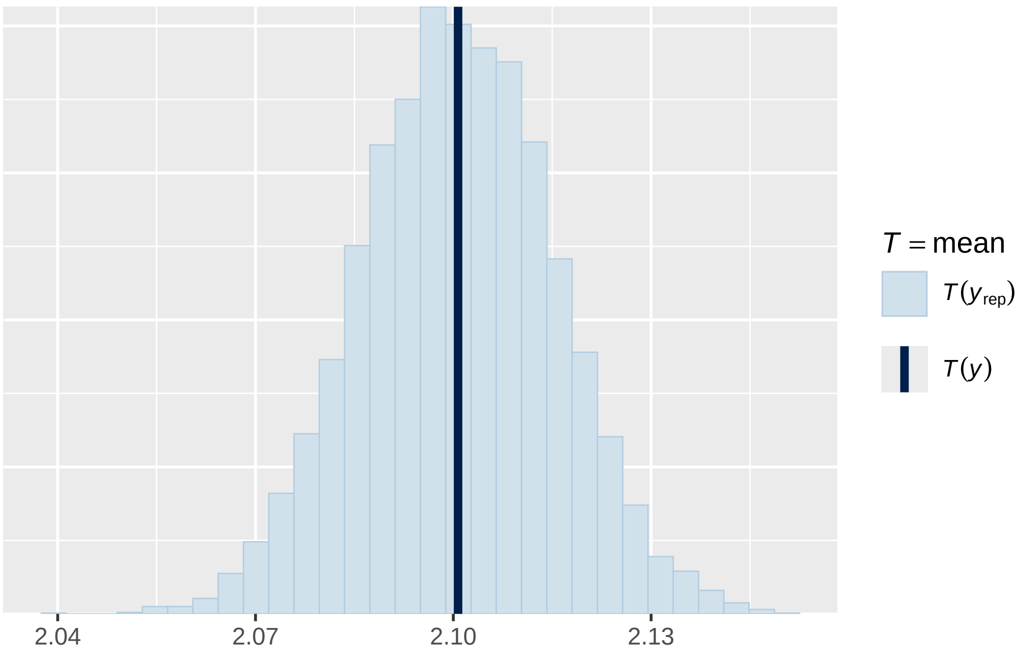
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



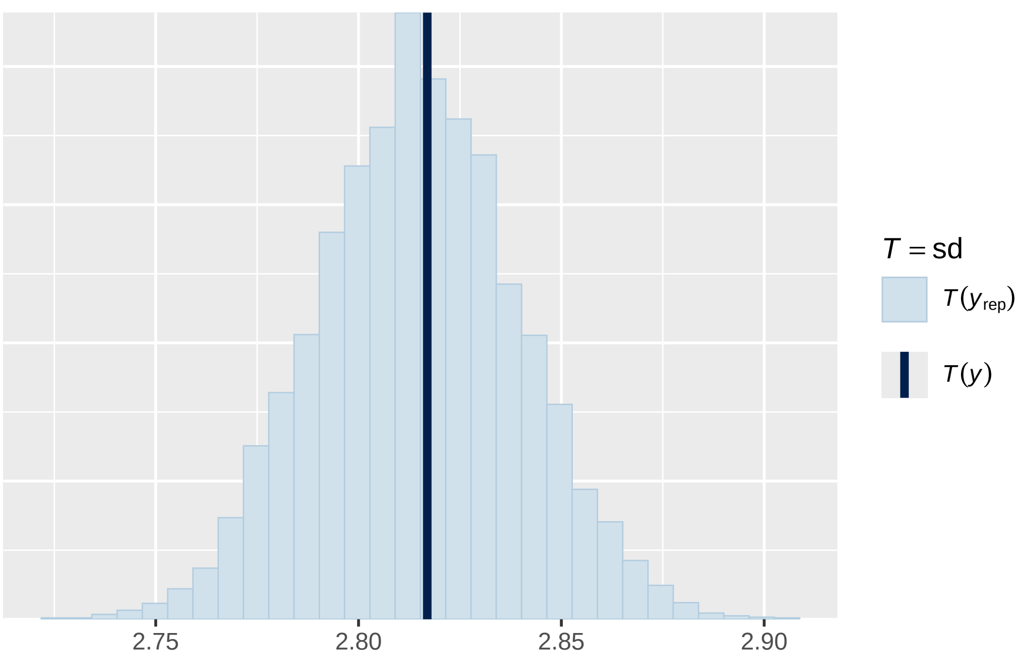
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

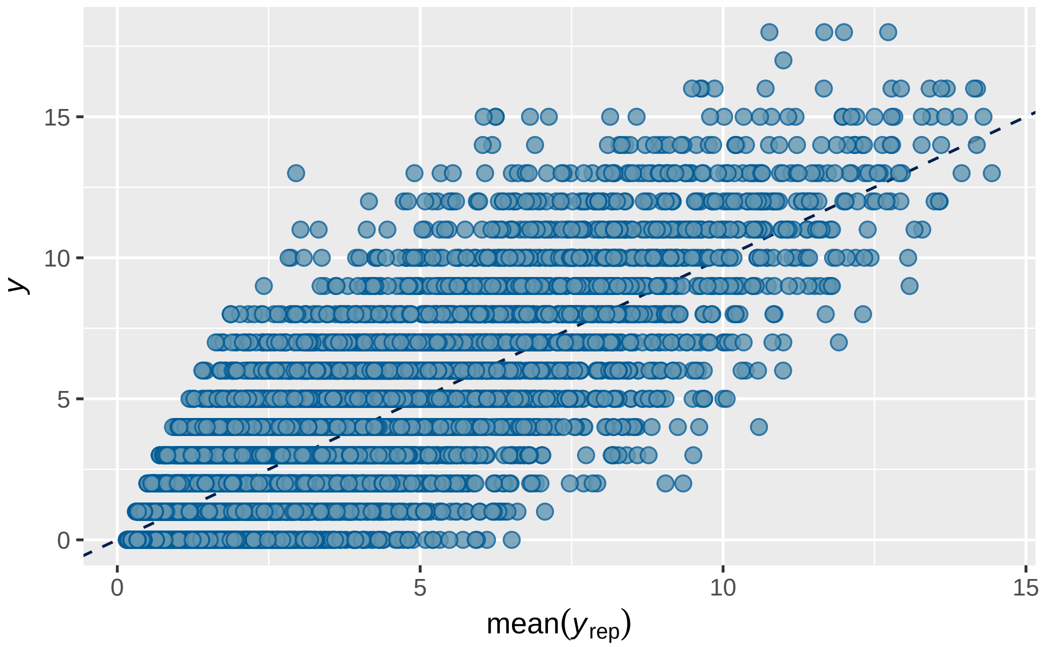


`stat_bin()` using `bins = 30`. Pick better value `binwidth`.




```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```





Model 1b output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.703   0.00329 0.696 0.709


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + (1 | participan
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.834     0.046    0.740    0.920 1.005      584     1177

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.036     0.036    0.967    1.107 1.002      732     1751

~site_id (Number of levels: 22)
```

```
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.176     0.039    0.111    0.264 1.000     6010     5359

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.044     0.051   -0.055    0.146 1.000     6379
breast_development_p_sc    0.059     0.019    0.023    0.095 1.001    12977
age_sc                    -0.058     0.016   -0.089   -0.027 1.000    14206
ethnraceblack_nh           0.027     0.067   -0.105    0.156 1.000     8354
ethnracehispanic          -0.033     0.061   -0.154    0.087 1.000     9225
ethnraceother              0.048     0.067   -0.083    0.179 1.000     8624
inr_sc                    -0.076     0.022   -0.119   -0.032 1.000    16951
                        Tail_ESS
Intercept                   5766
breast_development_p_sc     6378
age_sc                      6212
ethnraceblack_nh            6155
ethnracehispanic            6030
ethnraceother               5726
inr_sc                      5781

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

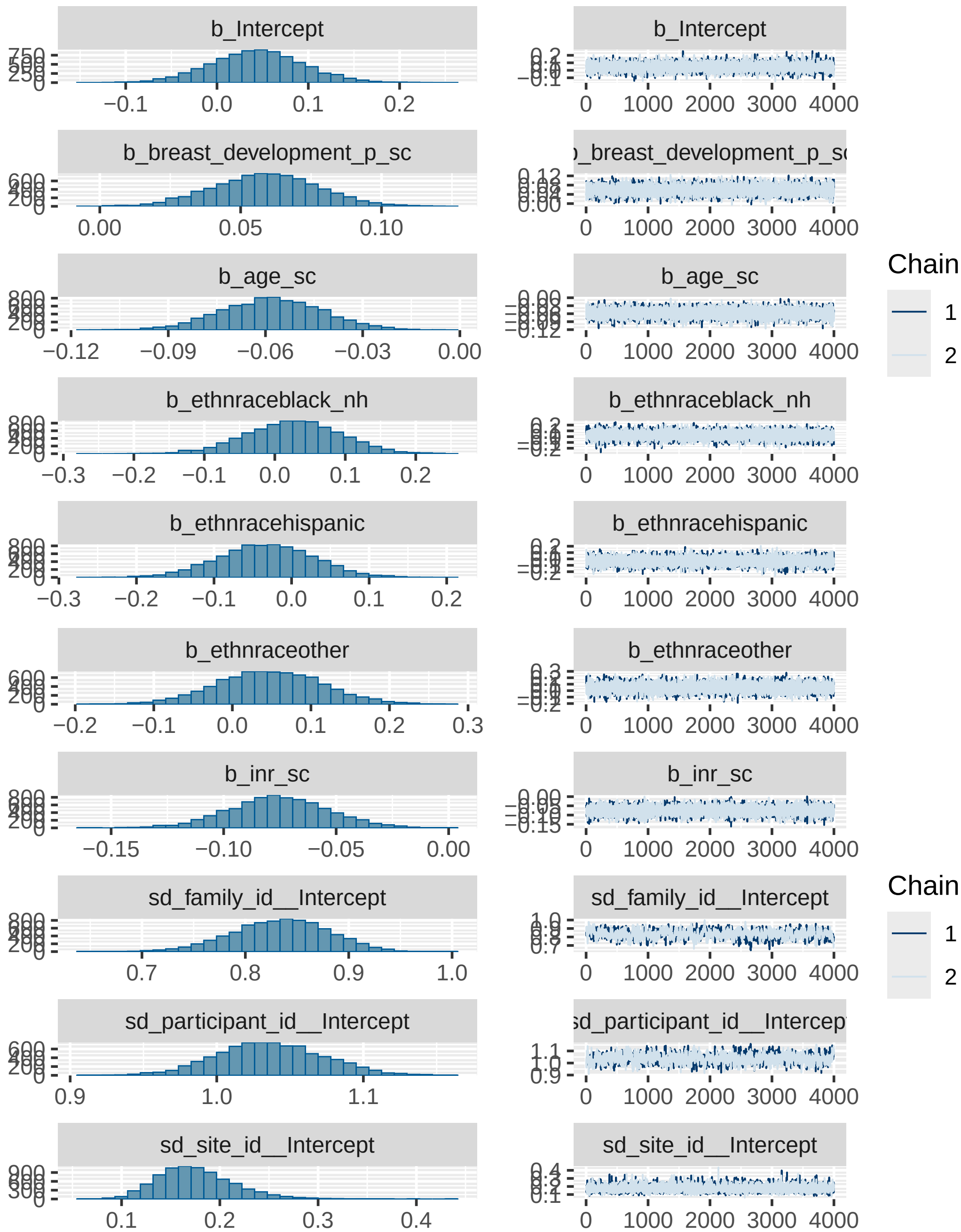
Model 1b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

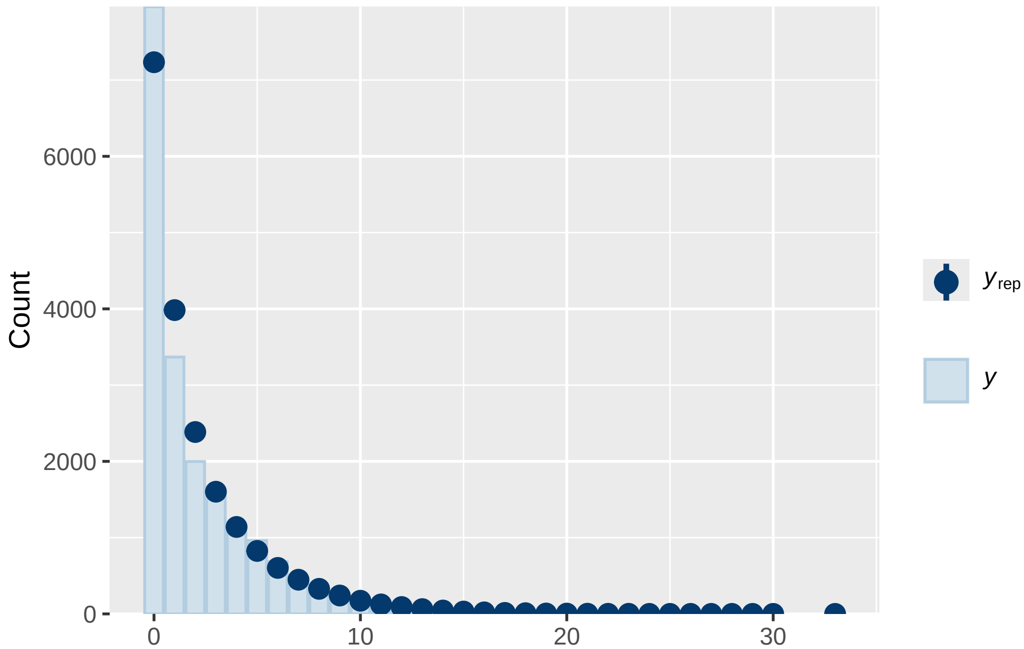
```
# A tibble: 1 x 4
  Variable           Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                 <dbl>          <dbl>          <dbl>
1 Breast development      3.7           1.43           6.05

[1] 0.812
```

Model 1b diagnostics

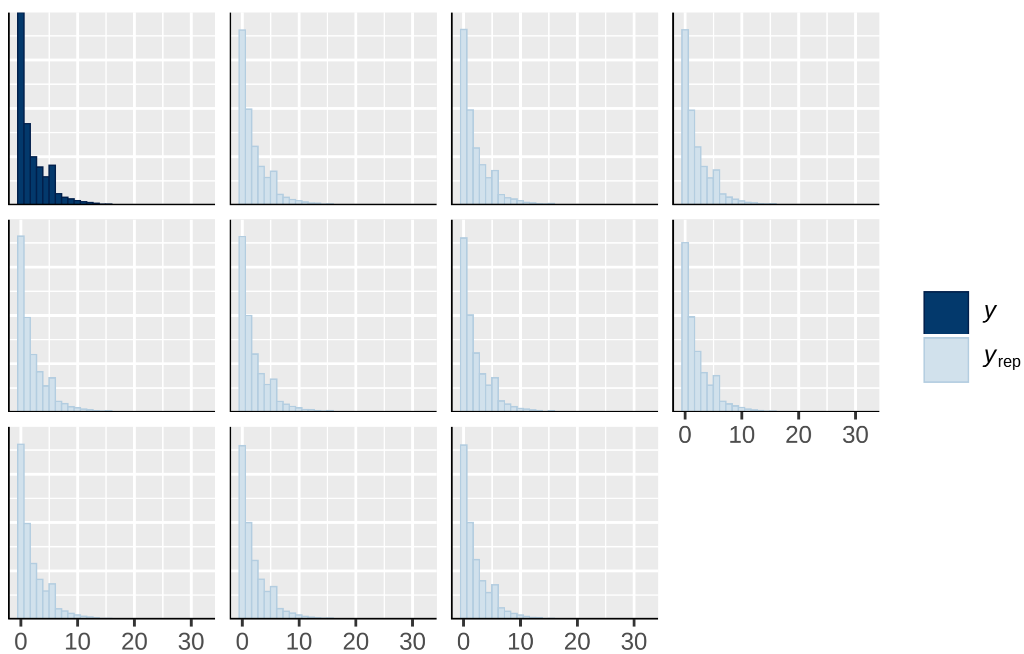
```
        prior     class                    coef          group resp
 normal(0, 1)         b
 normal(0, 1)         b                  age_sc
 normal(0, 1)         b breast_development_p_sc
```

```
        normal(0, 1)         b        ethnraceblack_nh
        normal(0, 1)         b        ethnracehispanic
        normal(0, 1)         b           ethnraceother
        normal(0, 1)         b                  inr_sc
student_t(3, 0, 2.7) Intercept
student_t(3, 0, 2.7)        sd
student_t(3, 0, 2.7)        sd                               family_id
student_t(3, 0, 2.7)        sd               Intercept      family_id
student_t(3, 0, 2.7)        sd                          participant_id
student_t(3, 0, 2.7)        sd               Intercept participant_id
student_t(3, 0, 2.7)        sd                                 site_id
student_t(3, 0, 2.7)        sd               Intercept         site_id
dpar nlpar lb ub tag       source
                             user
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                          default
            0             default
            0        (vectorized)
            0        (vectorized)
            0        (vectorized)
            0        (vectorized)
            0        (vectorized)
            0        (vectorized)
```

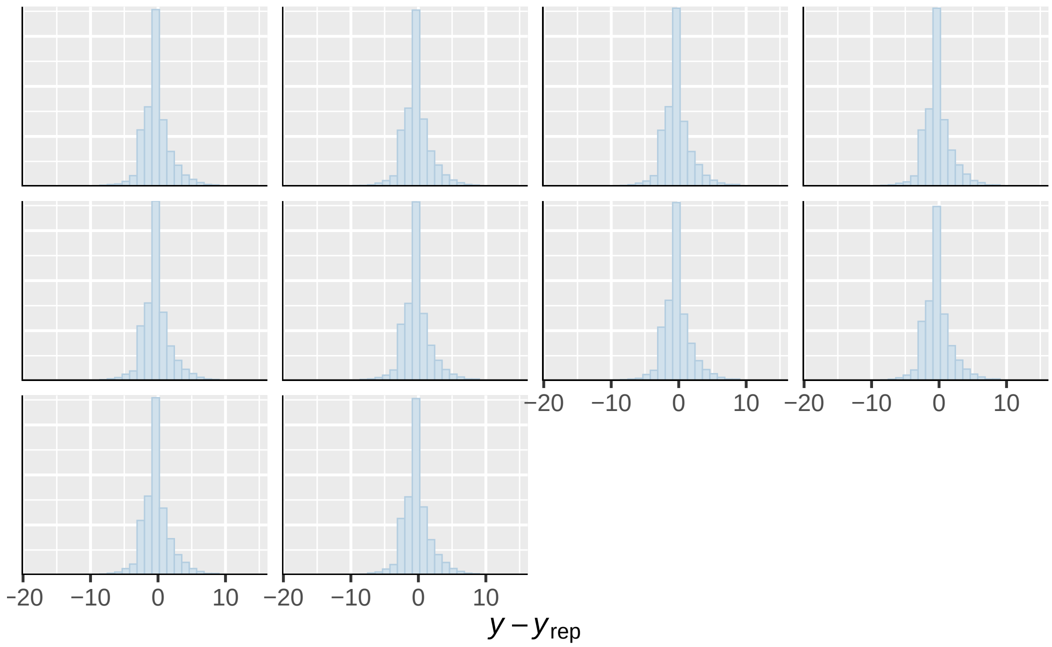

b_Intercept
−0.1 0.0 0.1 0.2
b_Intercept
0 1000 2000 3000 4000
b_breast_development_p_sc
0.00 0.05 0.10
b_age_sc
−0.12 −0.09 −0.06 −0.03 0.00
b_age_sc
b_ethnraceblack_nh
−0.3 −0.2 −0.1 0.0 0.1 0.2
b_ethnraceblack_nh
b_ethnracehispanic
−0.3 −0.2 −0.1 0.0 0.1 0.2
b_ethnracehispanic
b_ethnraceother
−0.2 −0.1 0.0 0.1 0.2 0.3
b_ethnraceother
b_inr_sc
−0.15 −0.10 −0.05 0.00
b_inr_sc
sd_family_id__Intercept
0.7 0.8 0.9 1.0
sd_family_id__Intercept
sd_participant_id__Intercept
0.9 1.0 1.1
sd_site_id__Intercept
0.1 0.2 0.3 0.4
sd_site_id__Intercept
Chain
1
2
Chain
1
2



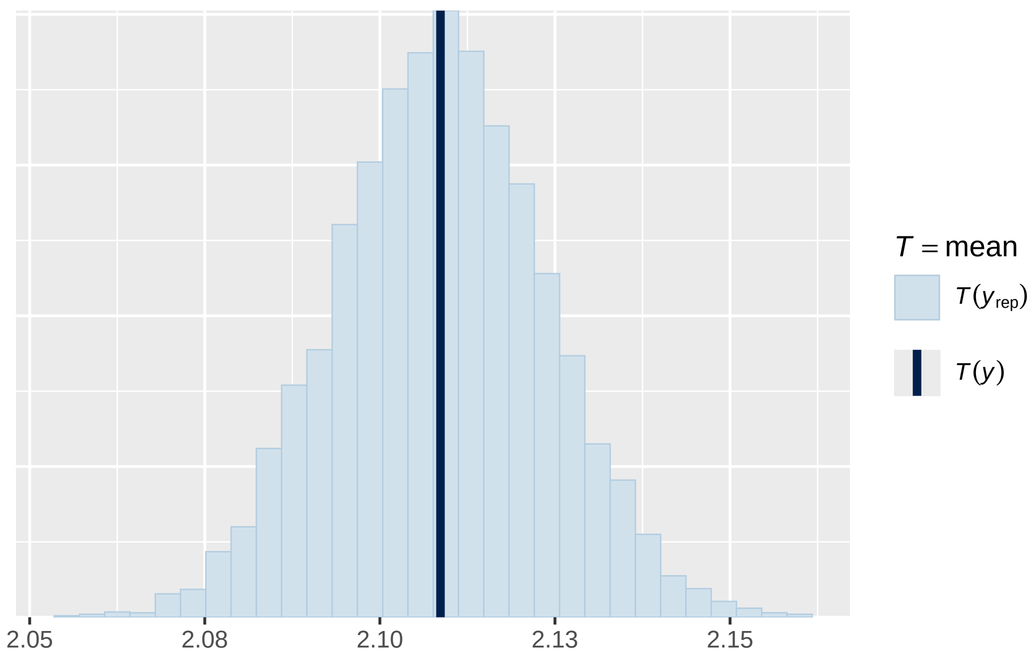
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



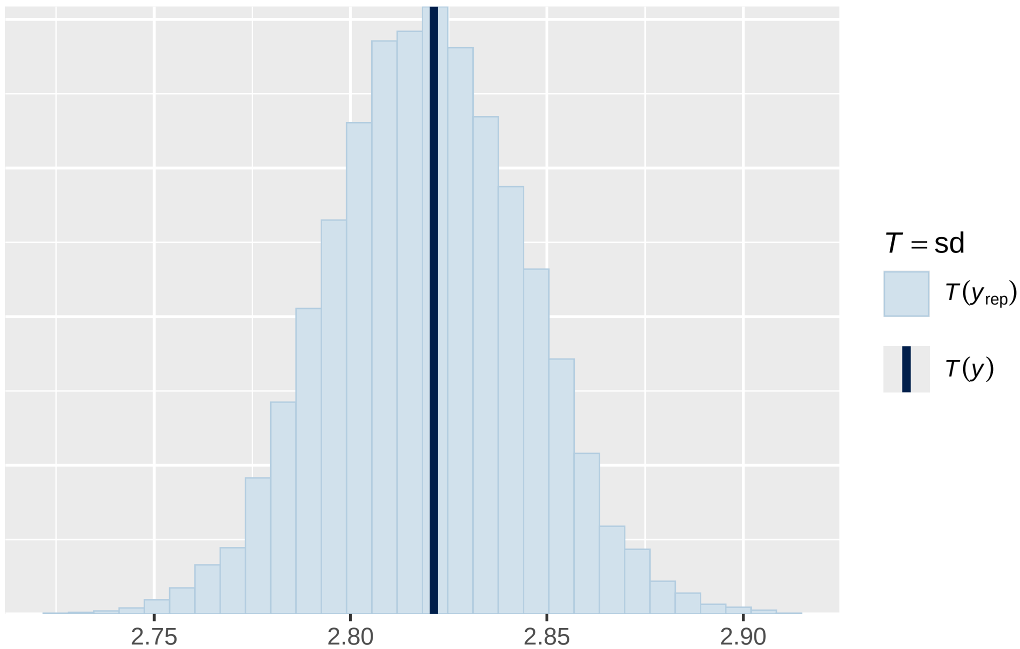
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

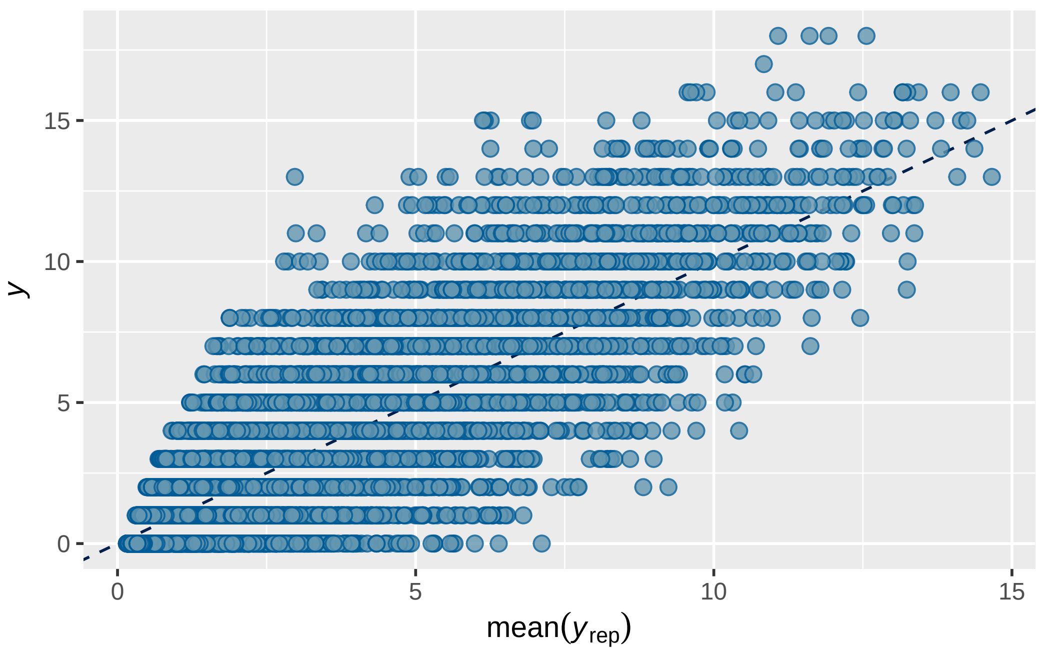


`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



`stat_bin()` using `bins = 30`. Pick better value `binwidth`.





Model 2 output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.839   0.00252 0.833 0.843
```

```
 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_status_life + age_sc + ethnrace + inr_sc +
   Data: analytic_df (Number of observations: 19375)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4620)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.761     0.018    0.726    0.795 1.002      926     1709

~obs_id (Number of levels: 19375)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.447     0.006    0.434    0.459 1.002     1555     3061

~participant_id (Number of levels: 5291)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.450     0.022    0.408    0.494 1.002      491     1018

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.119     0.027    0.074    0.181 1.001     3041     2635

Regression Coefficients:
                                    Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                              1.069     0.035    0.999    1.138 1.001
menarche_status_pY                     0.085     0.019    0.048    0.120 1.001
adhd_status_lifeY                      0.630     0.035    0.560    0.700 1.001
age_sc                                 0.053     0.015    0.023    0.083 1.000
ethnraceblack_nh                      -0.401     0.046   -0.490   -0.311 1.000
ethnracehispanic                      -0.070     0.042   -0.151    0.013 1.000
ethnraceother                         -0.018     0.045   -0.106    0.073 1.001
inr_sc                                -0.048     0.021   -0.089   -0.008 1.000
menarche_status_pY:adhd_status_lifeY  -0.020     0.031   -0.079    0.040 1.000
                                    Bulk_ESS Tail_ESS
Intercept                               2999     2686
menarche_status_pY                      4768     3184
adhd_status_lifeY                       3793     3257
age_sc                                  5096     3064
ethnraceblack_nh                        4020     3029
ethnracehispanic                        3863     2661
ethnraceother                           4413     3143
```

```
inr_sc                                   5848     3116
menarche_status_pY:adhd_status_lifeY     5492     3315

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

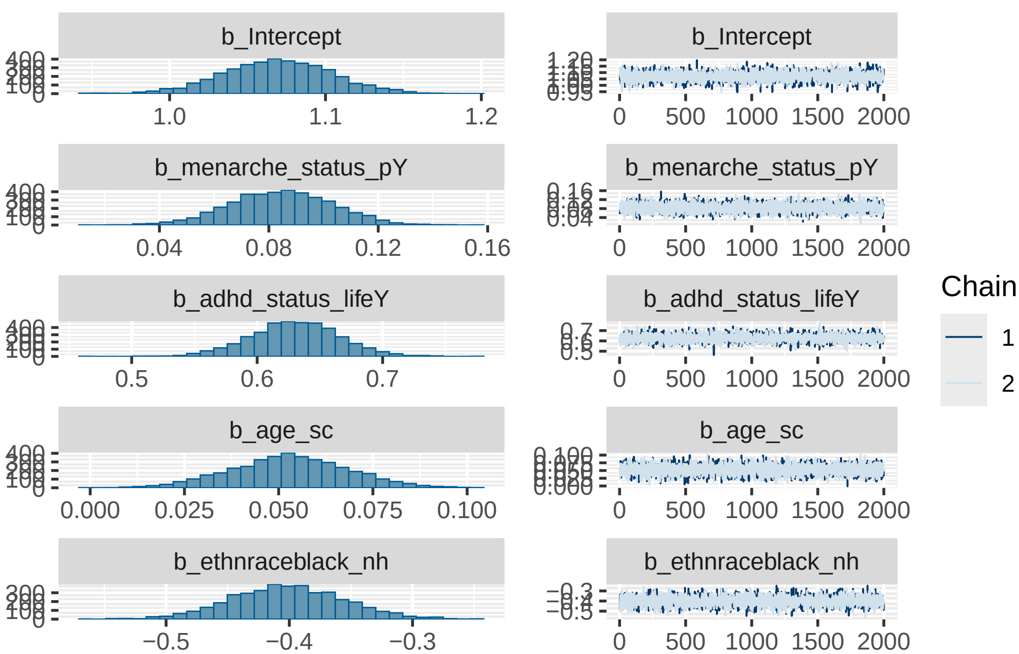
Model 2 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

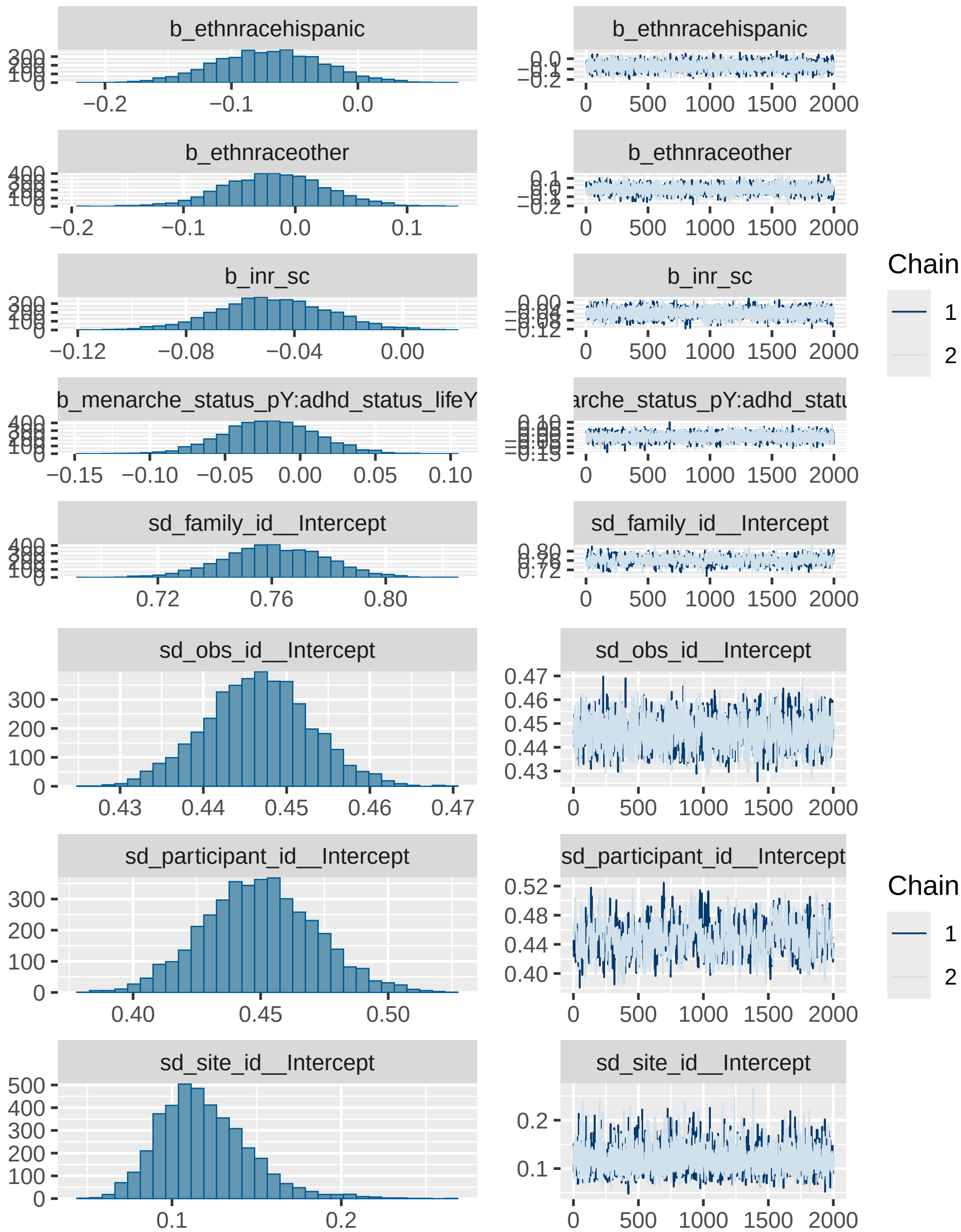
```
# A tibble: 3 x 4
  Variable                        Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                              <dbl>          <dbl>          <dbl>
1 Menarche (yes vs no) [ADHD = 0]     8.82           4.92          12.7
2 ADHD (yes vs no) [Menarche = 0]    87.8           75.1          101.
3 Interaction (Menarche × ADHD)      -2             -7.6             4.06
```

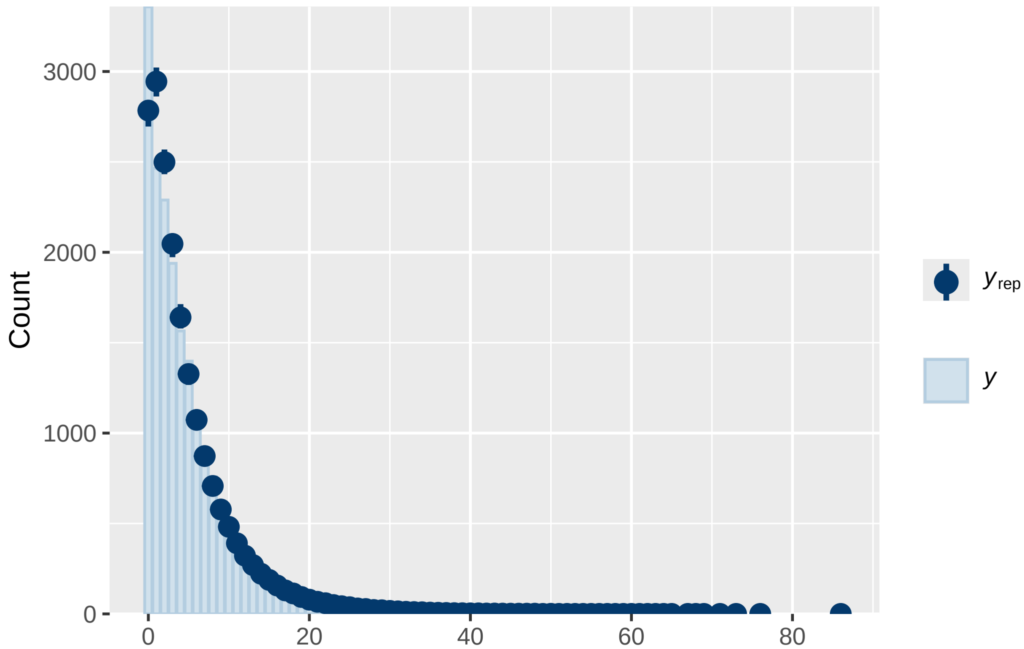
Model 2 diagnostics

```
                  prior     class                                coef
           normal(0, 1)         b
           normal(0, 1)         b                    adhd_status_lifeY
           normal(0, 1)         b                               age_sc
           normal(0, 1)         b                     ethnraceblack_nh
           normal(0, 1)         b                     ethnracehispanic
           normal(0, 1)         b                        ethnraceother
           normal(0, 1)         b                               inr_sc
           normal(0, 1)         b                   menarche_status_pY
           normal(0, 1)         b menarche_status_pY:adhd_status_lifeY
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                            Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                            Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                            Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                            Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
```

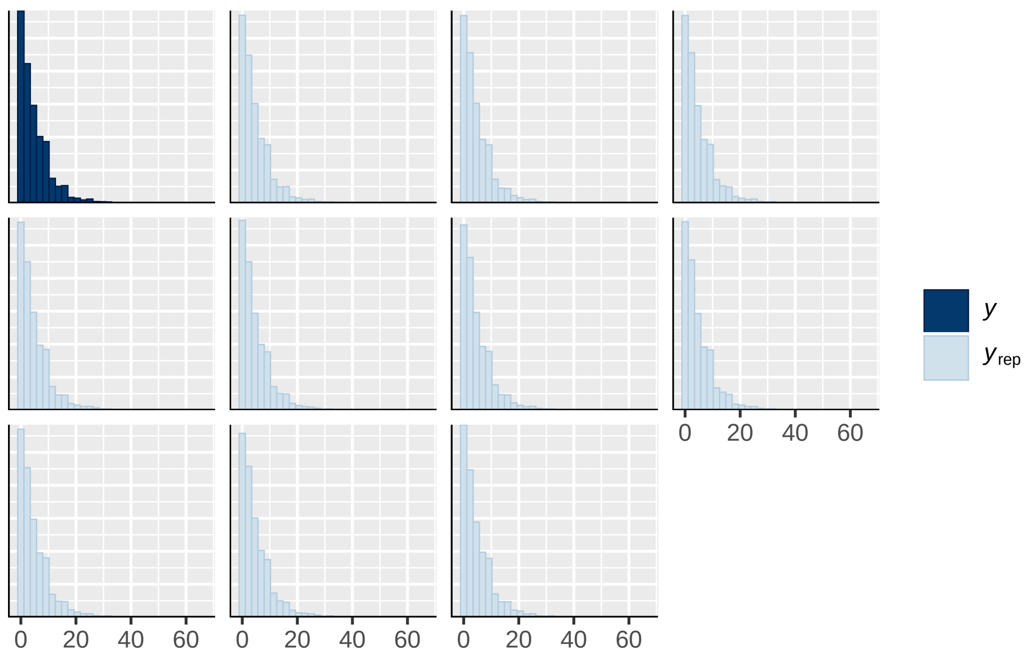
```
                                                 (vectorized)
                                                 (vectorized)
                                                 (vectorized)
                                                 (vectorized)
                                                 (vectorized)
                                                 (vectorized)
                                                 (vectorized)
                                                 (vectorized)
                                                      default
                                  0                   default
      family_id                   0              (vectorized)
      family_id                   0              (vectorized)
         obs_id                   0              (vectorized)
         obs_id                   0              (vectorized)
 participant_id                   0              (vectorized)
 participant_id                   0              (vectorized)
        site_id                   0              (vectorized)
        site_id                   0              (vectorized)
```



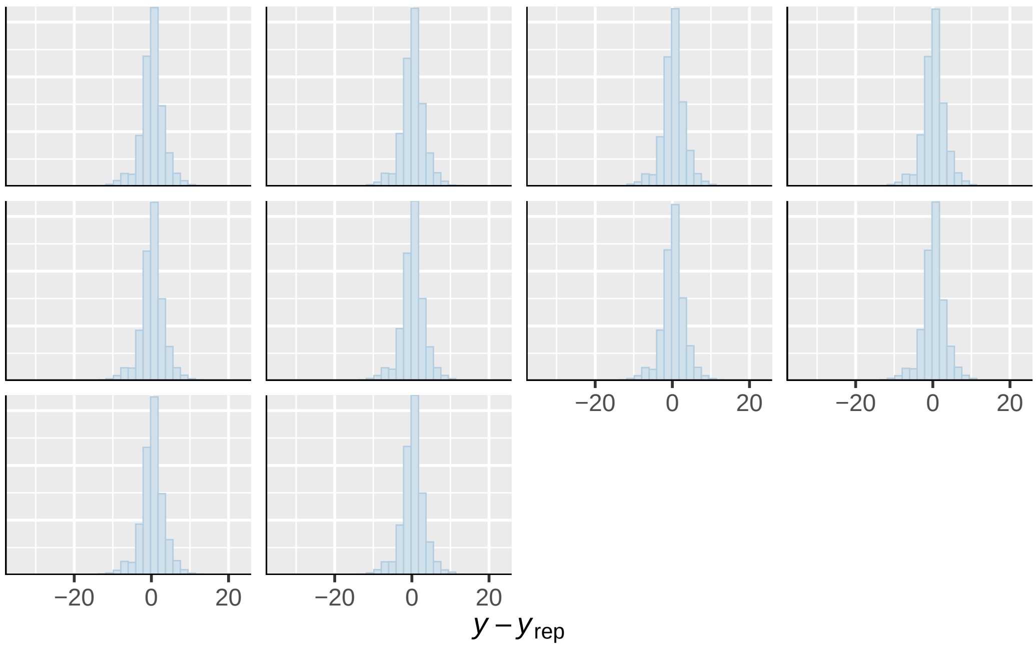

b_ethnracehispanic
b_ethnraceother
b_inr_sc
b_menarche_status_pY:adhd_status_lifeY
sd_family_id__Intercept
sd_obs_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2



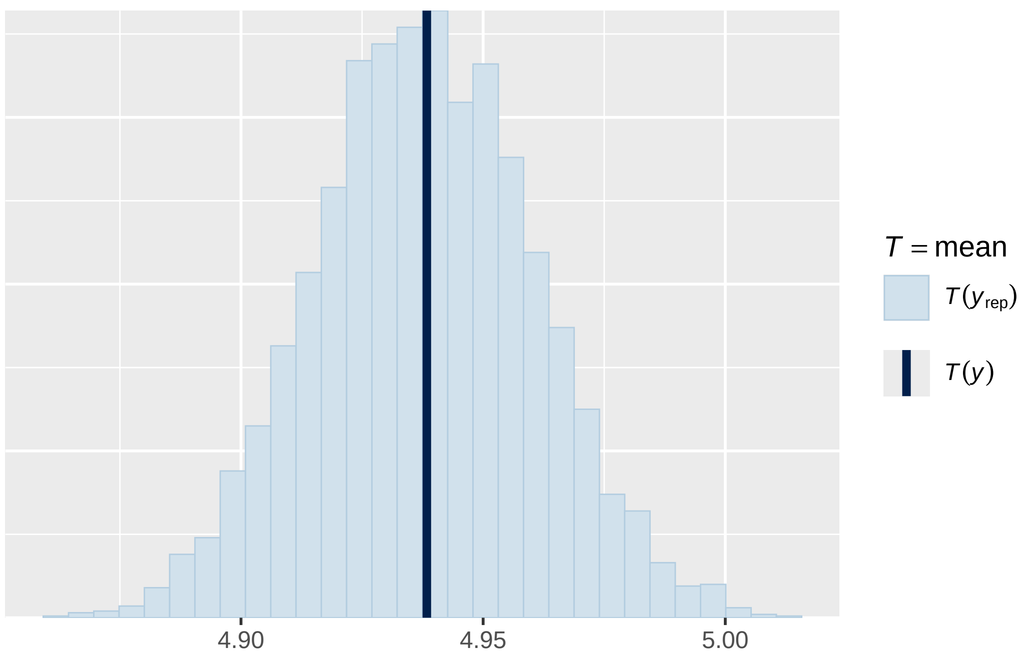
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



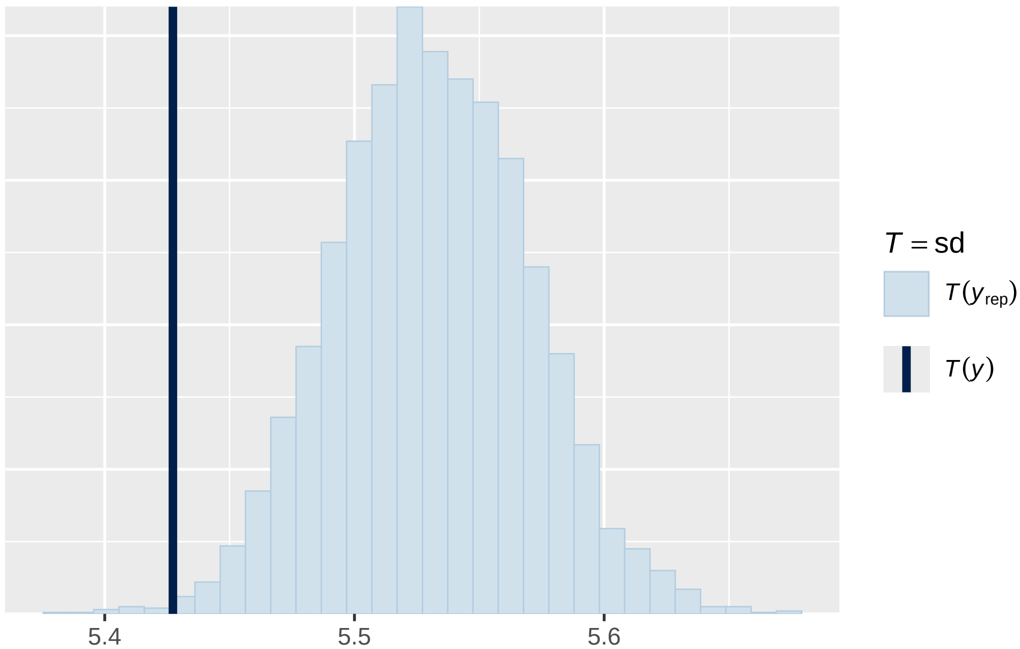
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

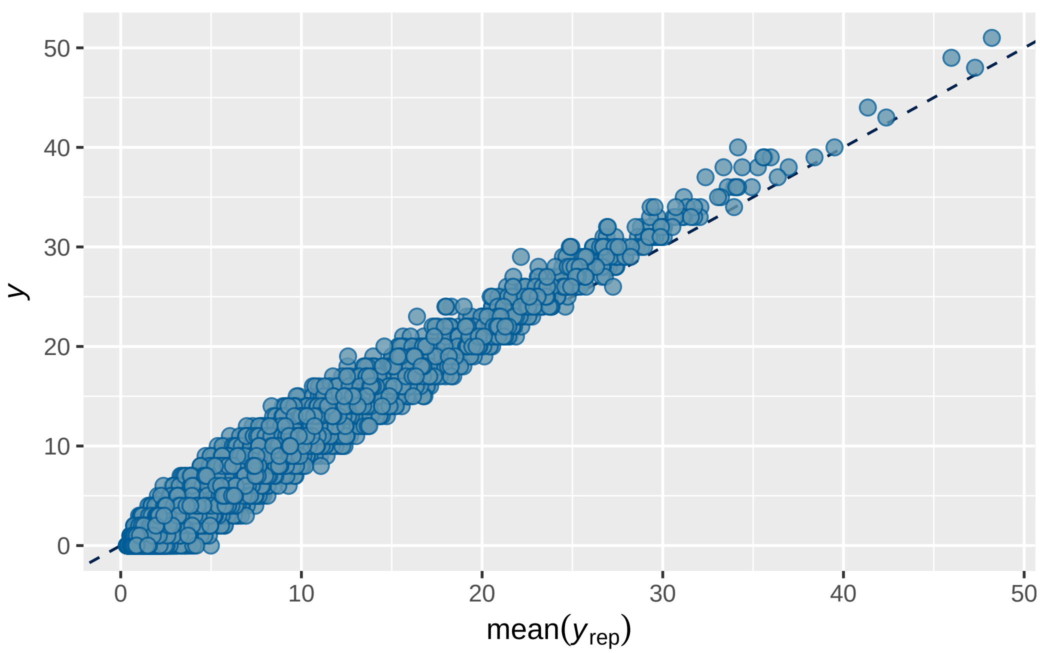



```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```





Model 2b output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.839   0.00252 0.834 0.844
```

```
 Family: poisson
  Links: mu = log
Formula: internalising ~ breast_development_p_sc * adhd_status_life + age_sc + ethnrace + in
   Data: analytic_df (Number of observations: 19358)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4613)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.760     0.018    0.724    0.794 1.002     1121     1823

~obs_id (Number of levels: 19358)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.446     0.006    0.434    0.459 1.000     3082     4811

~participant_id (Number of levels: 5283)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.450     0.022    0.407    0.493 1.002      647     1089

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.119     0.026    0.077    0.179 1.000     3098     4125

Regression Coefficients:
                                          Estimate Est.Error l-95% CI u-
95% CI
Intercept                                    1.094     0.034    1.025    1.164
breast_development_p_sc                      0.085     0.020    0.047    0.123
adhd_status_lifeY                            0.623     0.033    0.557    0.688
age_sc                                       0.058     0.015    0.028    0.088
ethnraceblack_nh                            -0.393     0.045   -0.483   -
0.305
ethnracehispanic                            -0.059     0.041   -0.140    0.020
ethnraceother                               -0.018     0.046   -0.106    0.072
inr_sc                                      -0.042     0.020   -0.081   -
0.002
breast_development_p_sc:adhd_status_lifeY   -0.037     0.033   -0.103    0.026
                                           Rhat Bulk_ESS Tail_ESS
Intercept                                 1.001     2867     3494
breast_development_p_sc                   1.001     4431     4816
adhd_status_lifeY                         1.000     3609     4548
age_sc                                    1.000     4709     4684
```

```
ethnraceblack_nh                              1.001     2975     4358
ethnracehispanic                              1.001     3858     4223
ethnraceother                                 1.001     3268     4697
inr_sc                                        1.001     4781     5121
breast_development_p_sc:adhd_status_lifeY 1.001     5083     5251

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

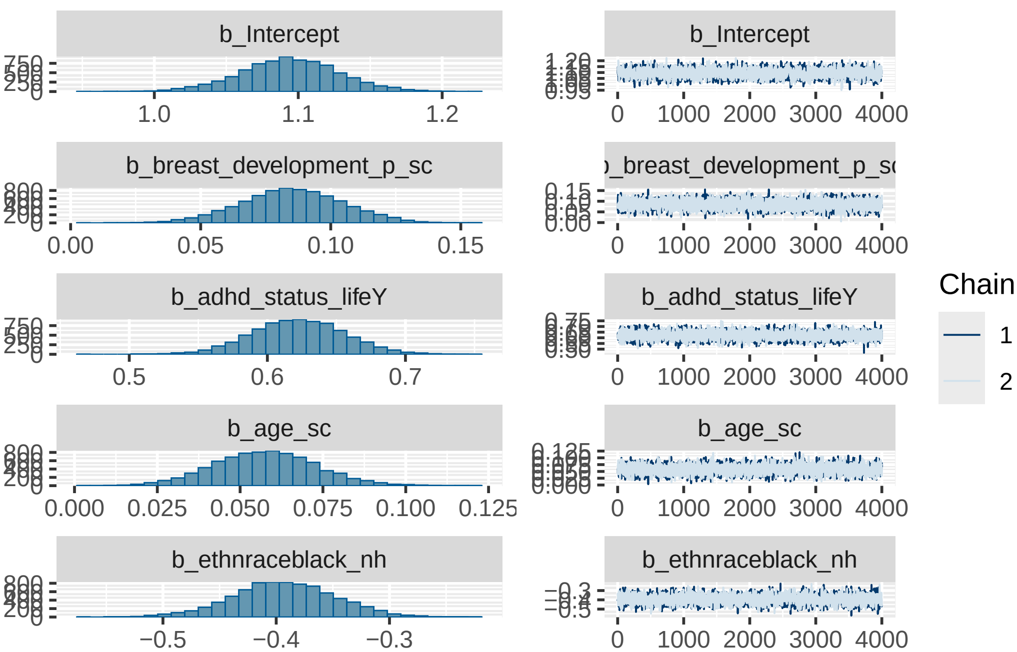
Model 2b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

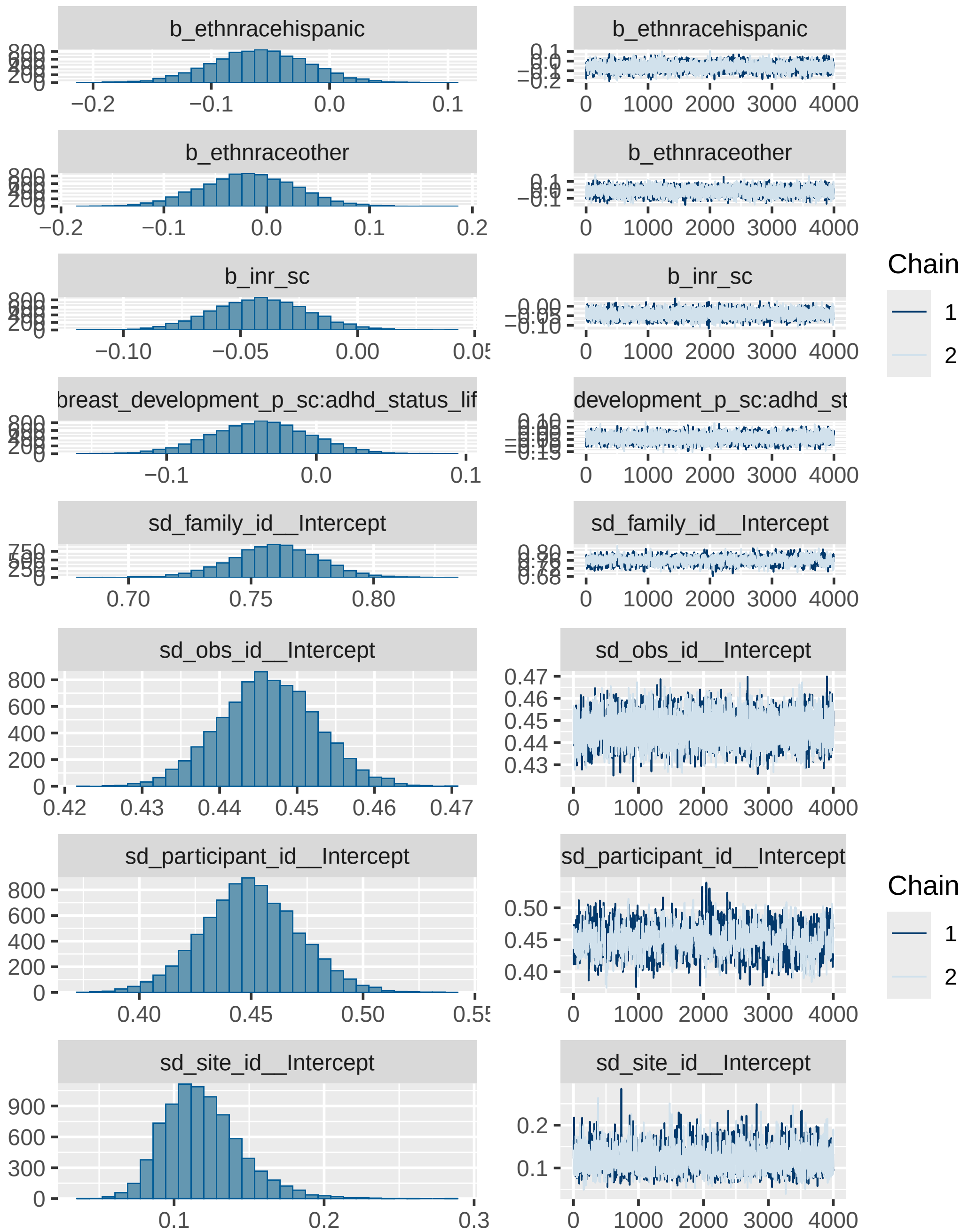
```
# A tibble: 3 x 4
  Variable                                   Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                         <dbl>          <dbl>          <dbl>
1 Breast development (per 1-unit) [ADHD ~      5.39           2.91           7.89
2 ADHD (yes vs no) [Breast development =~     86.4           74.6           98.9
3 Interaction (Breast development × ADHD)     -2.26          -6.15           1.63
```

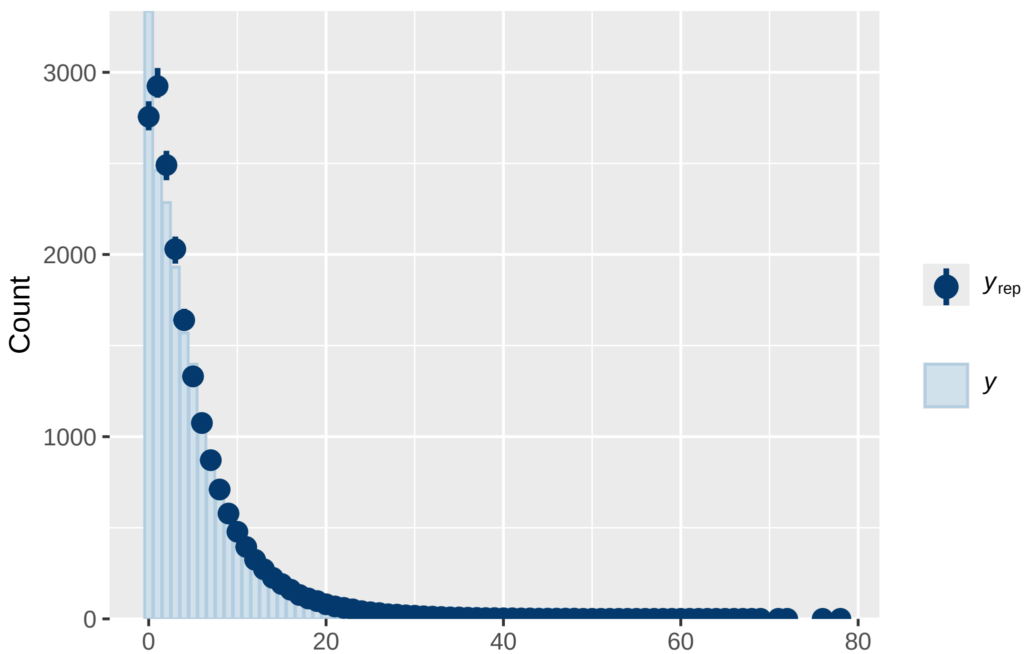
Model 2b diagnostics

```
                 prior     class                                      coef
          normal(0, 1)         b
          normal(0, 1)         b                         adhd_status_lifeY
          normal(0, 1)         b                                    age_sc
          normal(0, 1)         b                   breast_development_p_sc
          normal(0, 1)         b breast_development_p_sc:adhd_status_lifeY
          normal(0, 1)         b                           ethnraceblack_nh
          normal(0, 1)         b                          ethnracehispanic
          normal(0, 1)         b                             ethnraceother
          normal(0, 1)         b                                    inr_sc
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                 Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                 Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                 Intercept
   student_t(3, 0, 2.5)        sd
```

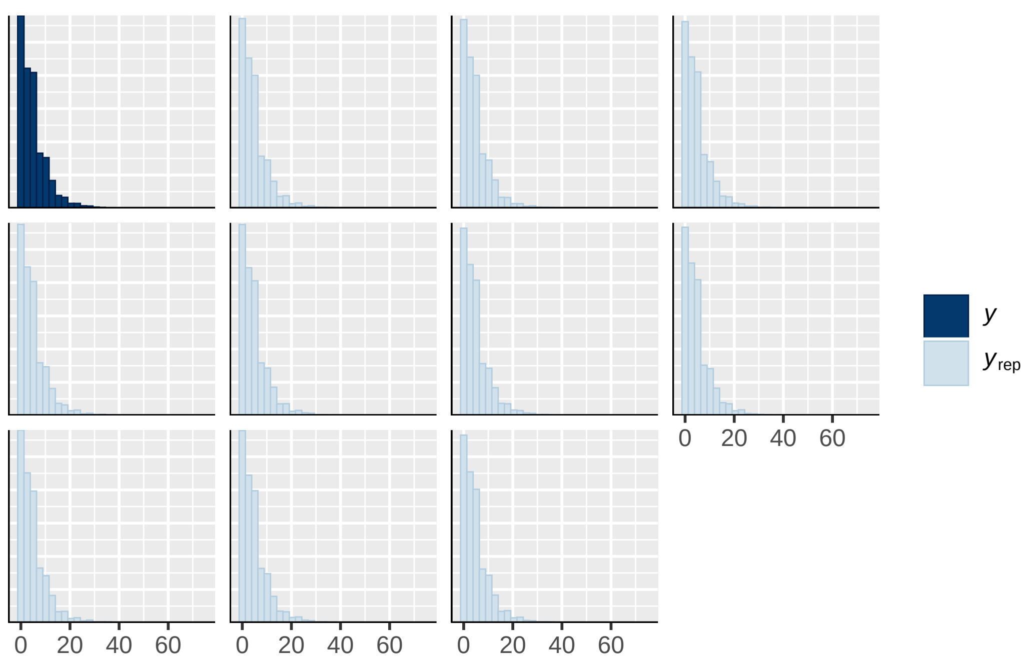
```
 student_t(3, 0, 2.5)        sd                                Intercept
        group resp dpar nlpar lb ub tag       source
                                                user
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                             default
                            0                default
   family_id                0           (vectorized)
   family_id                0           (vectorized)
      obs_id                0           (vectorized)
      obs_id                0           (vectorized)
participant_id              0           (vectorized)
participant_id              0           (vectorized)
     site_id                0           (vectorized)
     site_id                0           (vectorized)
```



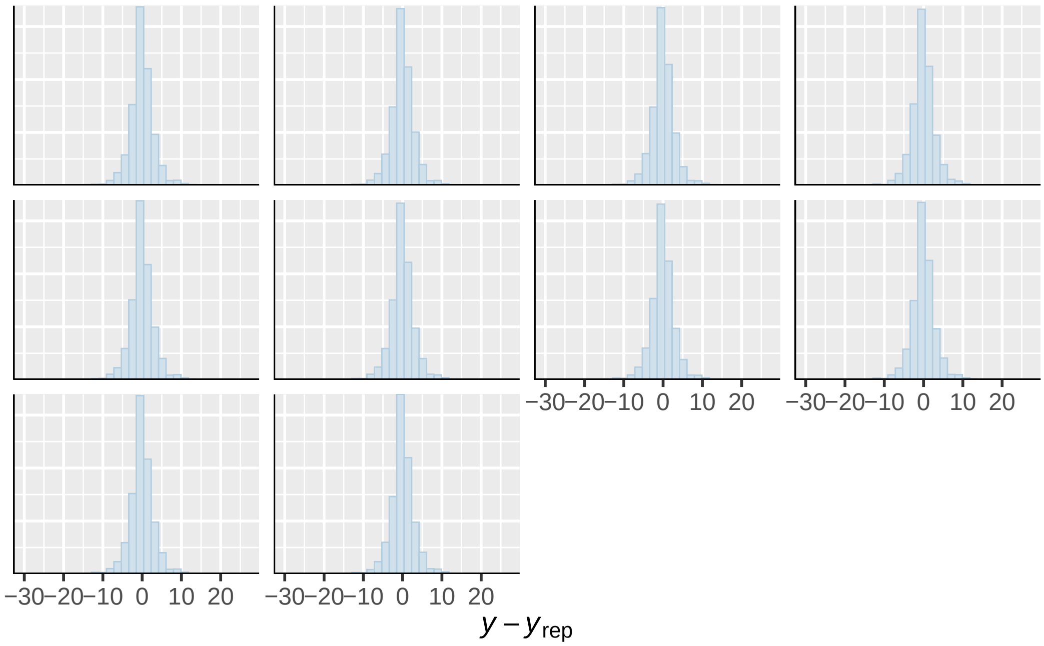

b_ethnracehispanic
b_ethnraceother
b_inr_sc
breast_development_p_sc:adhd_status_lif
development_p_sc:adhd_st
sd_family_id__Intercept
sd_obs_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2



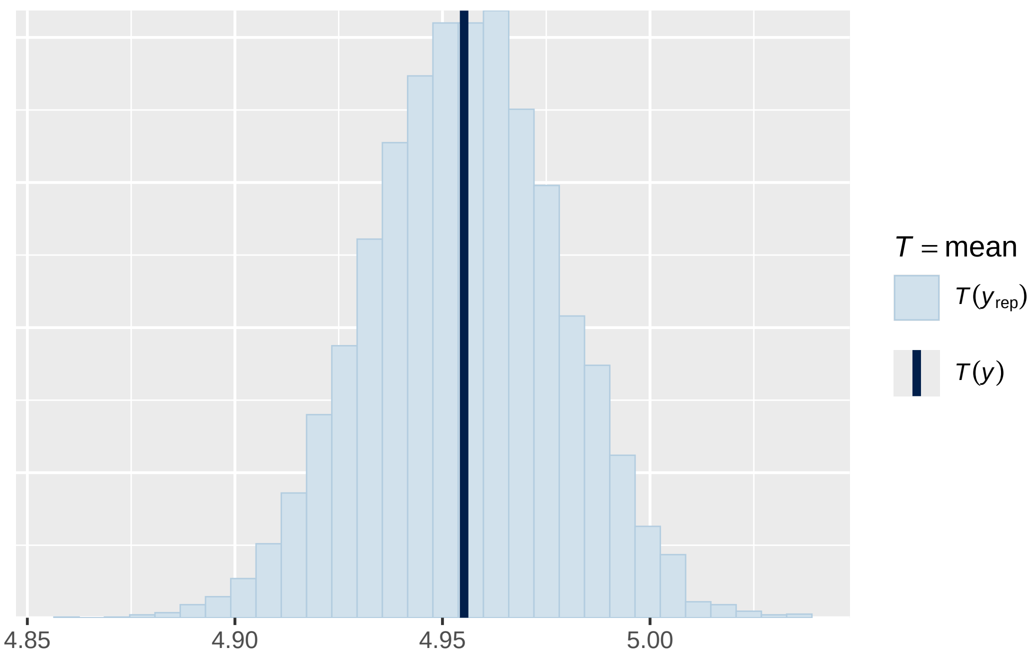
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



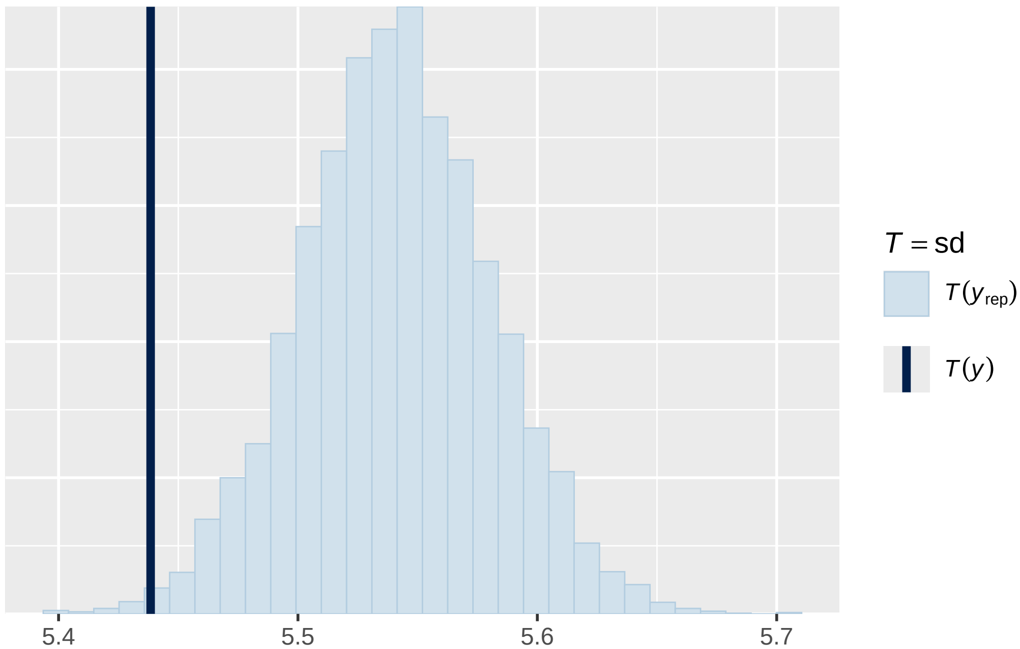
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

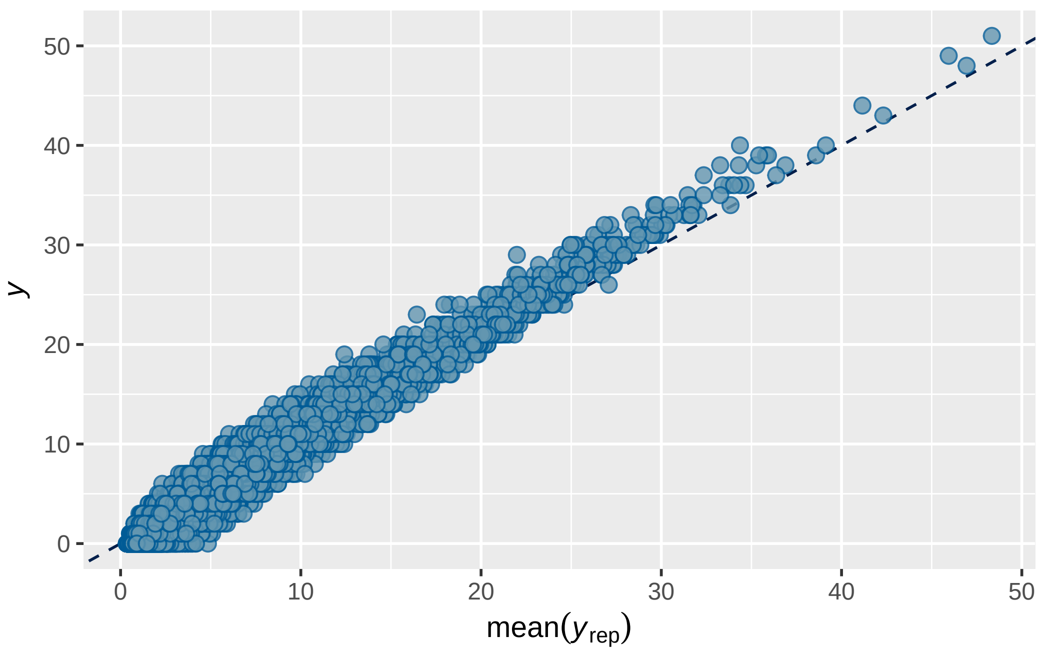



```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```





Model 3 output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.859   0.00254 0.854 0.864
```

```
 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_status_life + age_sc + ethnrace + inr_sc +
   Data: analytic_df (Number of observations: 19375)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4620)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.866     0.031    0.804    0.925 1.001      571      981

~obs_id (Number of levels: 19375)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.497     0.008    0.481    0.513 1.001     1635     2749

~participant_id (Number of levels: 5291)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.767     0.030    0.708    0.827 1.001      482      929

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.154     0.034    0.097    0.229 1.000     2806     2777

Regression Coefficients:
                                   Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                             0.117     0.046    0.029    0.208 1.000
menarche_status_pY                    0.104     0.024    0.056    0.150 1.000
adhd_status_lifeY                     1.337     0.046    1.250    1.428 1.000
age_sc                               -0.133     0.020   -0.171   -0.095 1.000
ethnraceblack_nh                     -0.005     0.058   -0.117    0.108 1.000
ethnracehispanic                     -0.011     0.054   -0.118    0.096 1.000
ethnraceother                        -0.008     0.060   -0.126    0.108 1.000
inr_sc                               -0.185     0.026   -0.237   -0.133 1.000
menarche_status_pY:adhd_status_lifeY  -0.038     0.037   -0.110    0.035 1.000
                                   Bulk_ESS Tail_ESS
Intercept                              2834     2840
menarche_status_pY                     4417     2993
adhd_status_lifeY                      3473     2959
age_sc                                 4313     3393
ethnraceblack_nh                       3811     3114
ethnracehispanic                       3719     3092
ethnraceother                          4016     3057
```

```
inr_sc                                    4859     2898
menarche_status_pY:adhd_status_lifeY     4283     2964

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

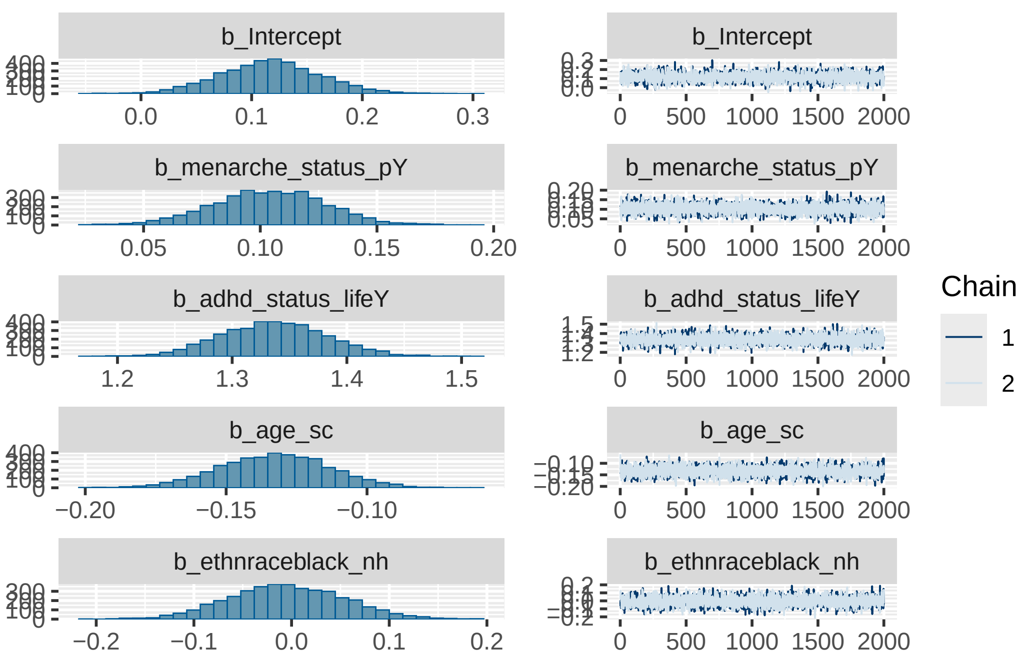
Model 3 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

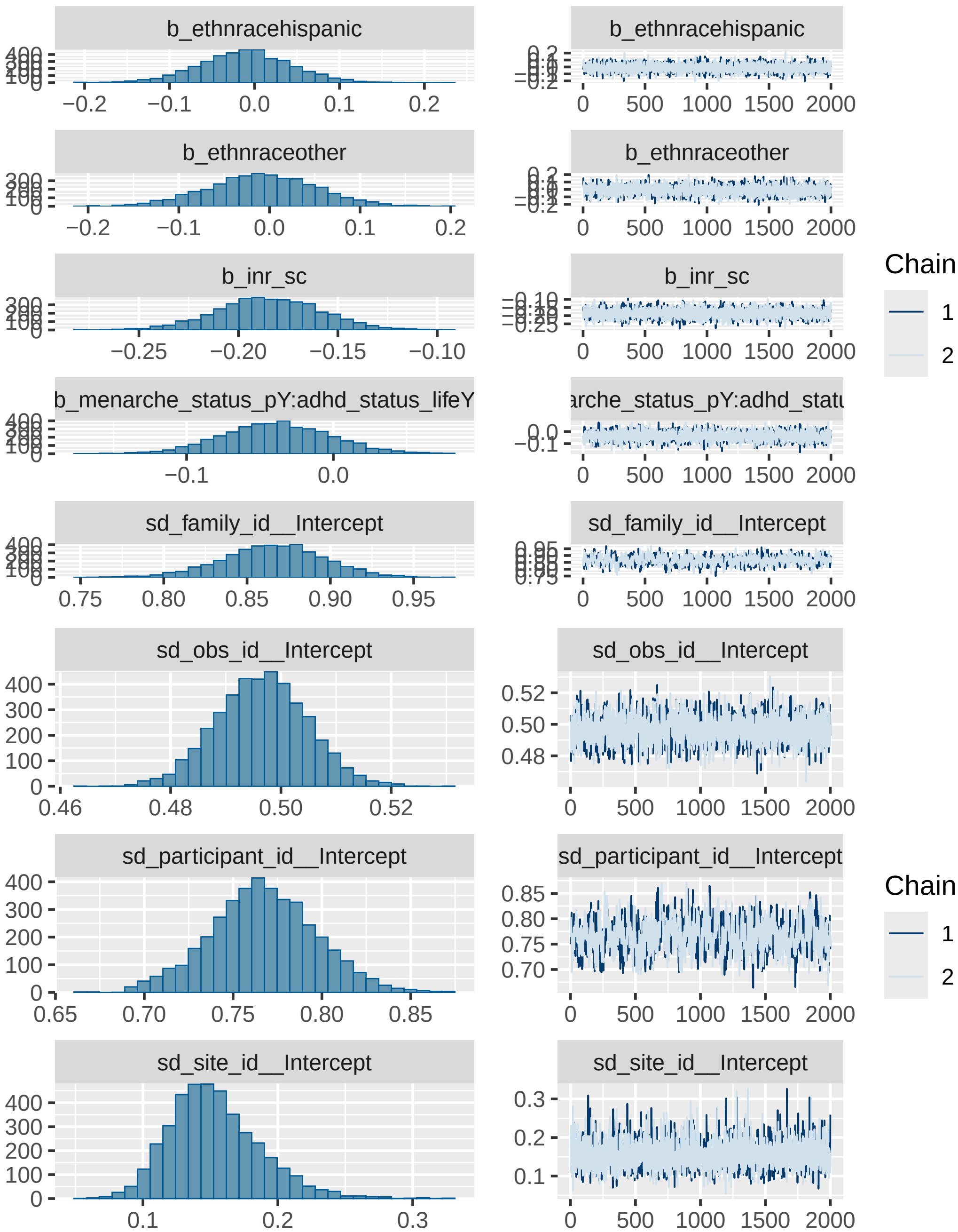
```
# A tibble: 3 x 4
  Variable                        Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                              <dbl>          <dbl>          <dbl>
1 Menarche (yes vs no) [ADHD = 0]    11.0            5.79           16.2
2 ADHD (yes vs no) [Menarche = 0]   281.           249.            317.
3 Interaction (Menarche × ADHD)      -3.78         -10.4             3.56
```

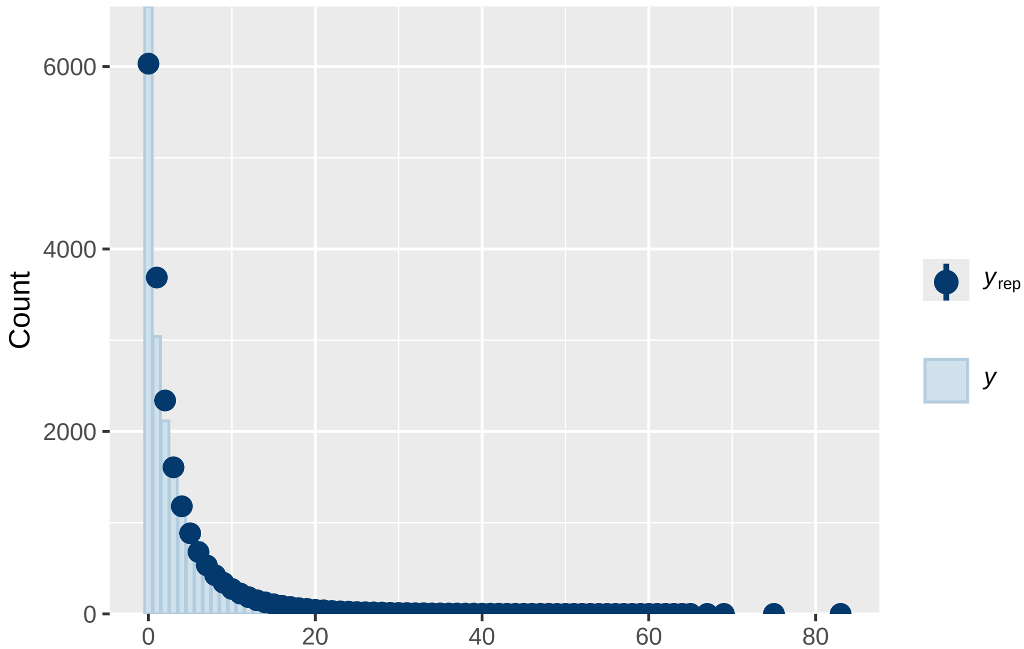
Model 3 diagnostics

```
                prior     class                                 coef
         normal(0, 1)         b
         normal(0, 1)         b                    adhd_status_lifeY
         normal(0, 1)         b                               age_sc
         normal(0, 1)         b                    ethnraceblack_nh
         normal(0, 1)         b                    ethnracehispanic
         normal(0, 1)         b                       ethnraceother
         normal(0, 1)         b                               inr_sc
         normal(0, 1)         b                  menarche_status_pY
         normal(0, 1)         b menarche_status_pY:adhd_status_lifeY
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                            Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                            Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                            Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                            Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
```

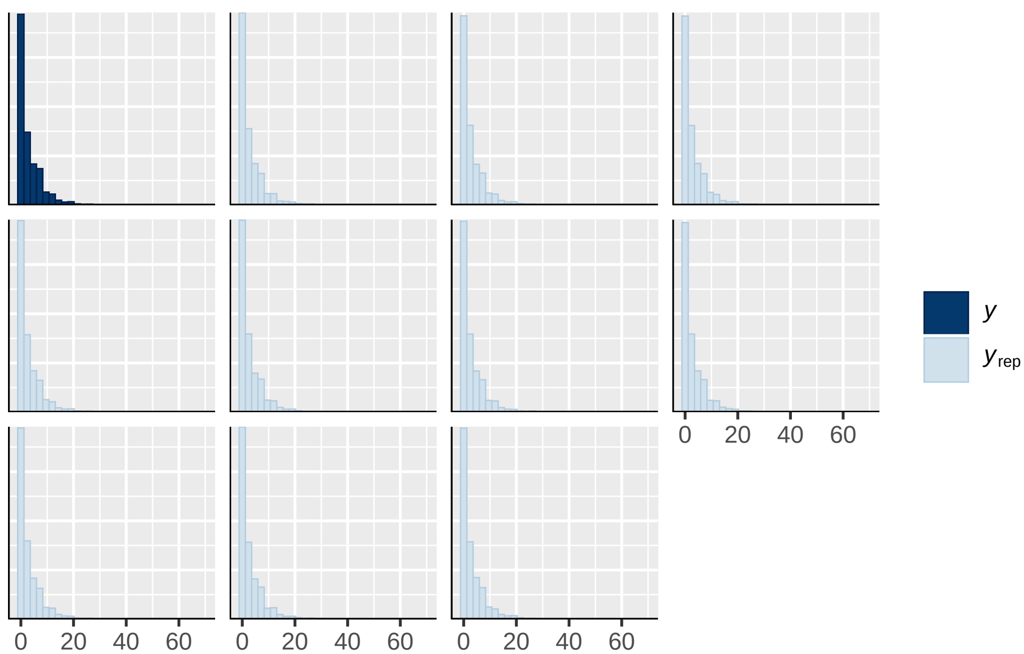
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
obs_id	0	(vectorized)
obs_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



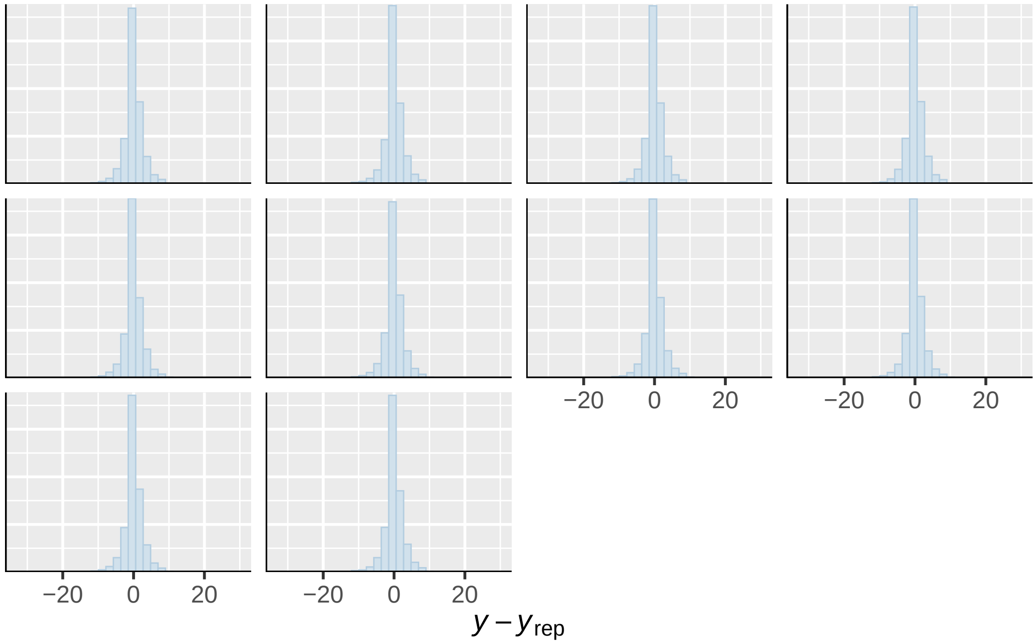

b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_sc
b_inr_sc
b_menarche_status_pY:adhd_status_lifeY
sd_family_id__Intercept
sd_family_id__Intercept
sd_obs_id__Intercept
sd_obs_id__Intercept
sd_participant_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
Chain
1
2
Chain
1
2



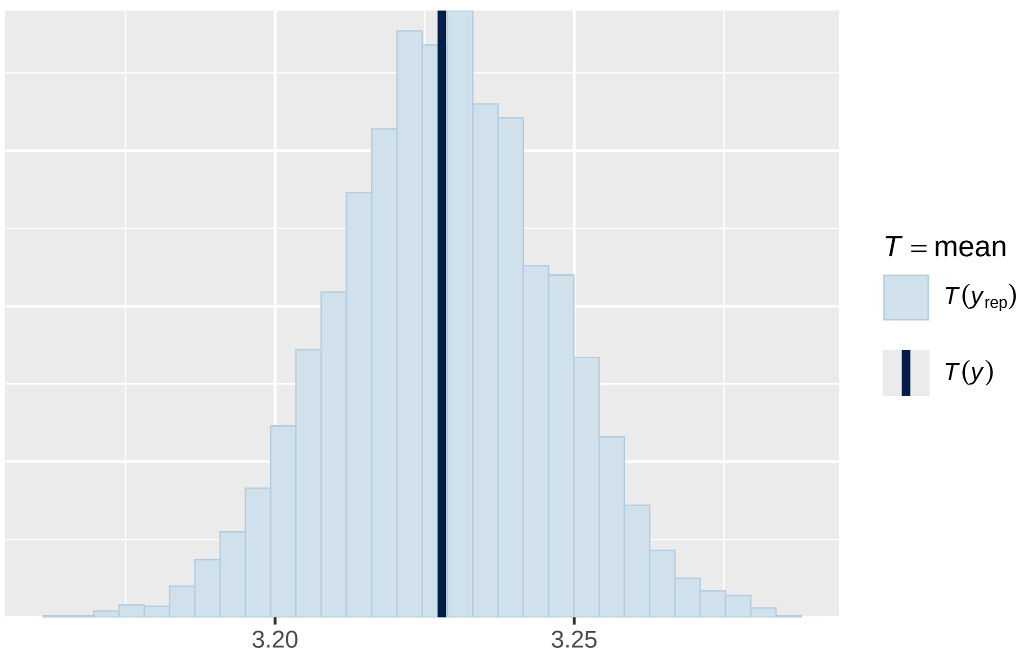

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



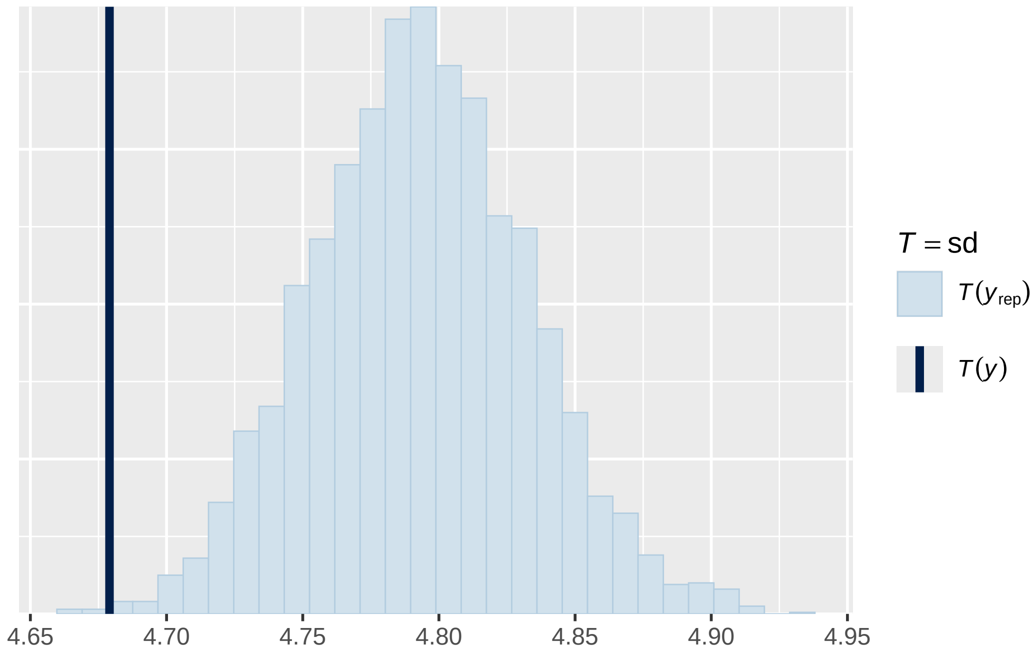

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



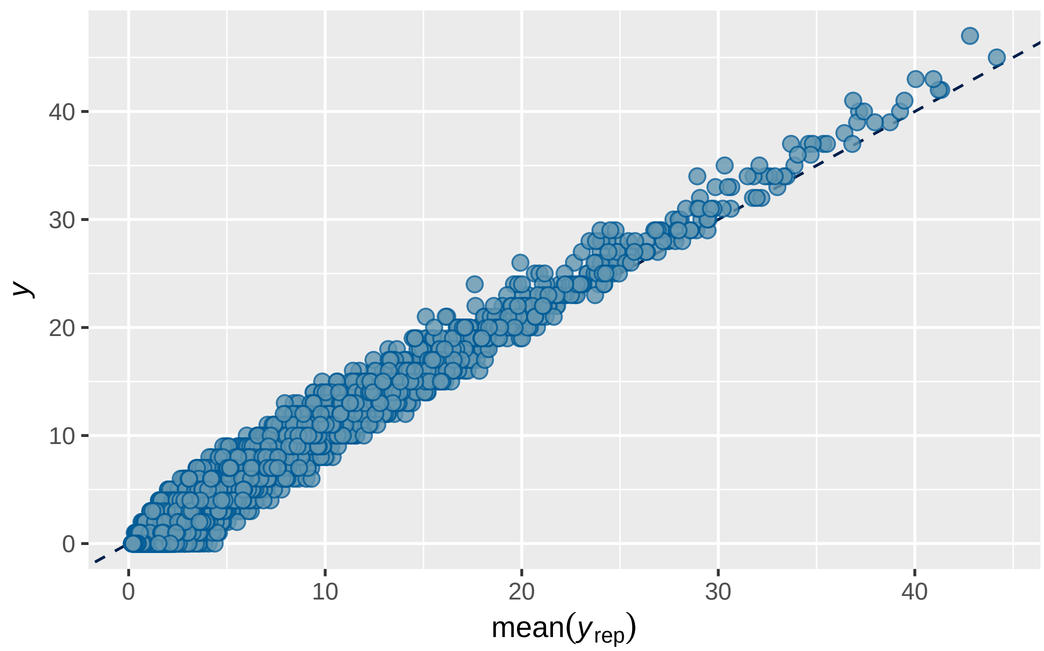
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



4.65
4.70
4.75
4.80
4.85
4.90
4.95
$T = \text{sd}$
$T(y_{\text{rep}})$
$T(y)$



40
30
20
10
0
y
0
10
20
30
40
$\text{mean}(y_{\text{rep}})$

Model 3b output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.859   0.00256 0.854 0.864


 Family: poisson
  Links: mu = log
Formula: externalising ~ breast_development_p_sc * adhd_status_life + age_sc + ethnrace + in
   Data: analytic_df (Number of observations: 19358)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4613)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.859     0.031    0.798    0.917 1.001      822     1858

~obs_id (Number of levels: 19358)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.494     0.008    0.478    0.510 1.001     3006     5123

~participant_id (Number of levels: 5283)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.769     0.031    0.710    0.830 1.001      732     1812

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.153     0.034    0.097    0.231 1.000     5190     5332

Regression Coefficients:
                                          Estimate Est.Error l-95% CI u-
95% CI
Intercept                                    0.156     0.045    0.066    0.245
breast_development_p_sc                      0.148     0.025    0.099    0.197
adhd_status_lifeY                            1.320     0.044    1.235    1.405
age_sc                                      -0.149     0.019   -0.186   -
0.112
ethnraceblack_nh                            -0.014     0.058   -0.125    0.102
ethnracehispanic                            -0.005     0.054   -0.110    0.101
ethnraceother                               -0.013     0.060   -0.129    0.104
inr_sc                                      -0.174     0.027   -0.226   -
0.122
```

```
breast_development_p_sc:adhd_status_lifeY   -0.064     0.037   -0.136    0.008
                                           Rhat Bulk_ESS Tail_ESS
Intercept                                 1.000     4898     5183
breast_development_p_sc                   1.000     7220     6679
adhd_status_lifeY                         1.000     7073     5690
age_sc                                    1.000     7978     6418
ethnraceblack_nh                          1.000     6482     6244
ethnracehispanic                          1.001     6871     5548
ethnraceother                             1.000     6827     5715
inr_sc                                    1.000     8890     6277
breast_development_p_sc:adhd_status_lifeY 1.000     7550     5964

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

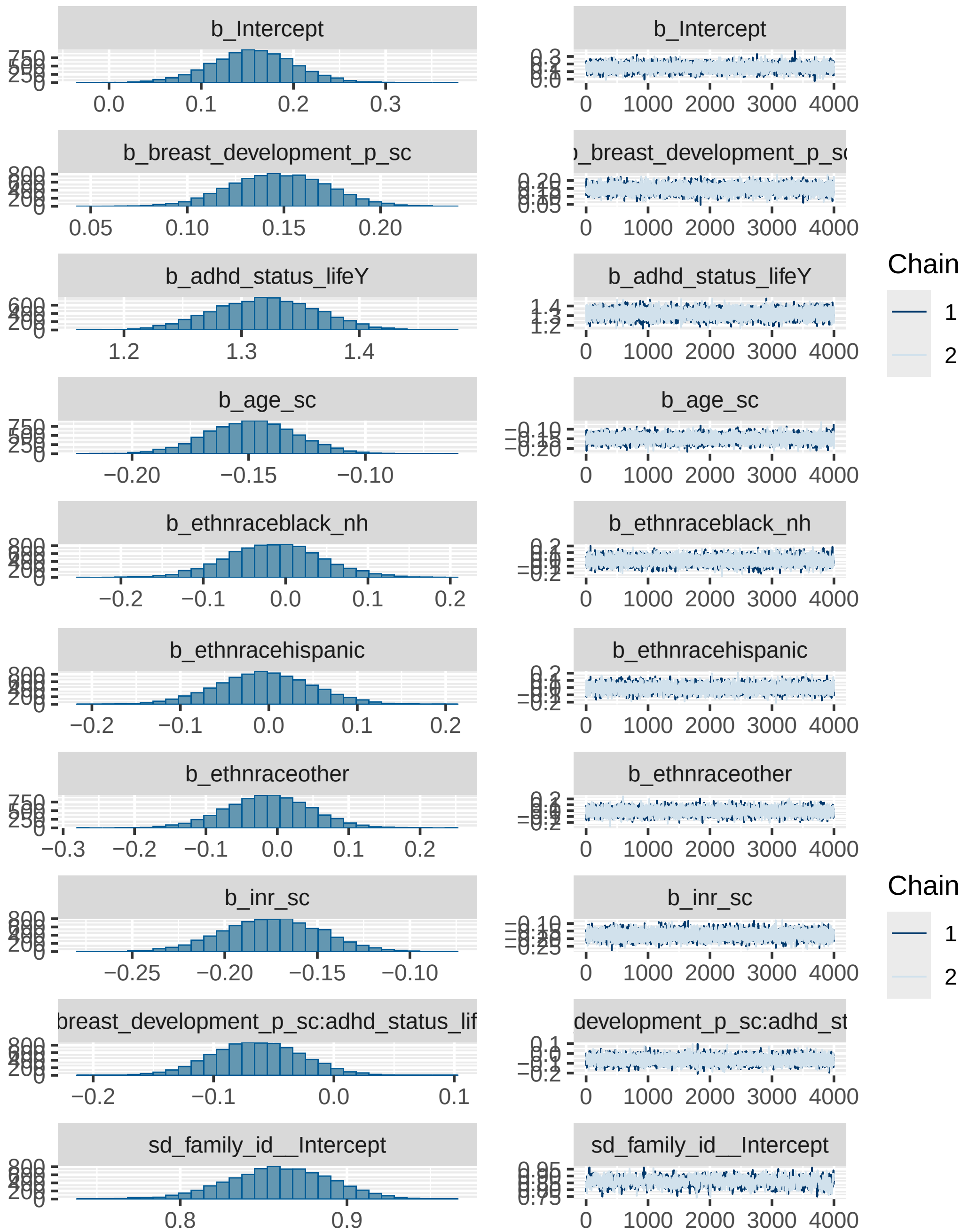
Model 3b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

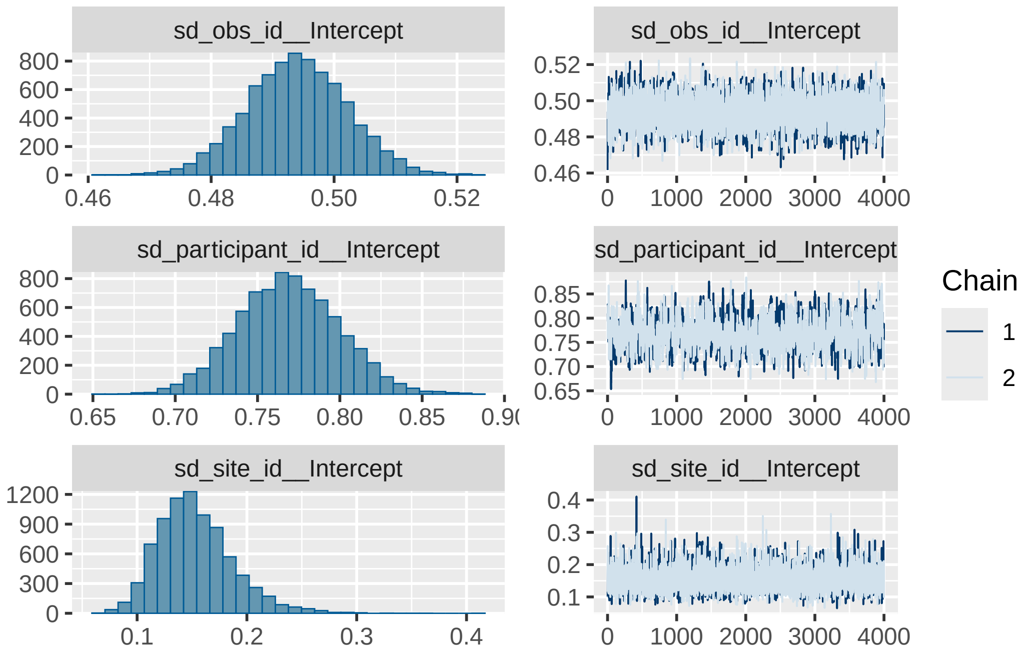
```
# A tibble: 3 x 4
  Variable                                 Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                       <dbl>          <dbl>          <dbl>
1 Breast development (per 1-unit) [ADHD ~      9.51            6.3           12.9
2 ADHD (yes vs no) [Breast development =~   274.             244.           307.
3 Interaction (Breast development × ADHD)    -3.86          -8.01           0.49
```

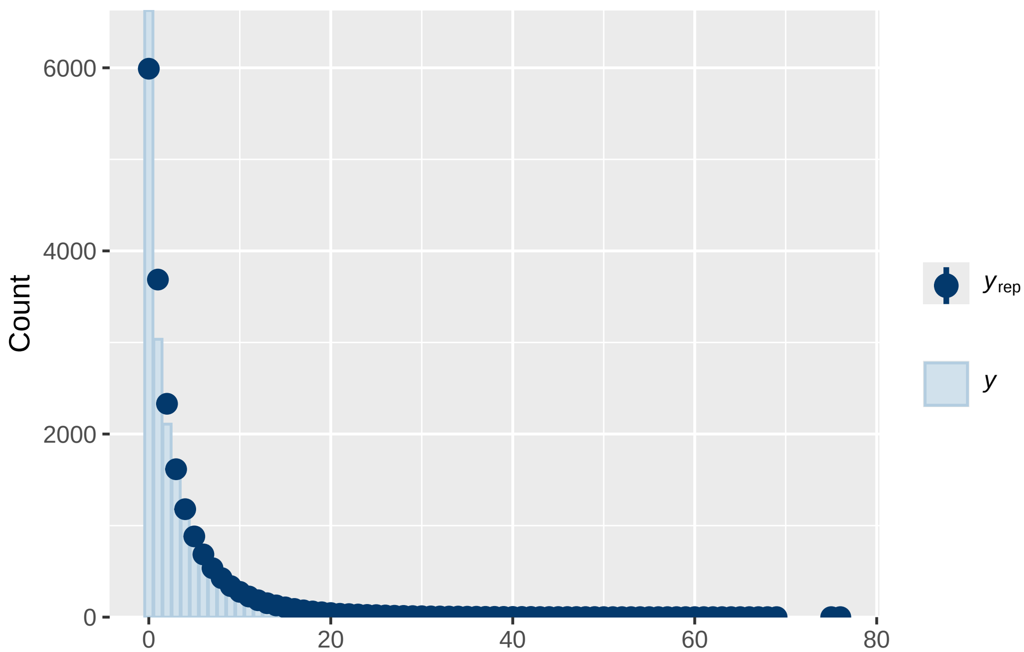
Model 3b diagnostics

```
                prior     class                                      coef
         normal(0, 1)         b
         normal(0, 1)         b                         adhd_status_lifeY
         normal(0, 1)         b                                    age_sc
         normal(0, 1)         b                   breast_development_p_sc
         normal(0, 1)         b breast_development_p_sc:adhd_status_lifeY
         normal(0, 1)         b                          ethnraceblack_nh
         normal(0, 1)         b                          ethnracehispanic
         normal(0, 1)         b                             ethnraceother
         normal(0, 1)         b                                    inr_sc
 student_t(3, 0.7, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
```

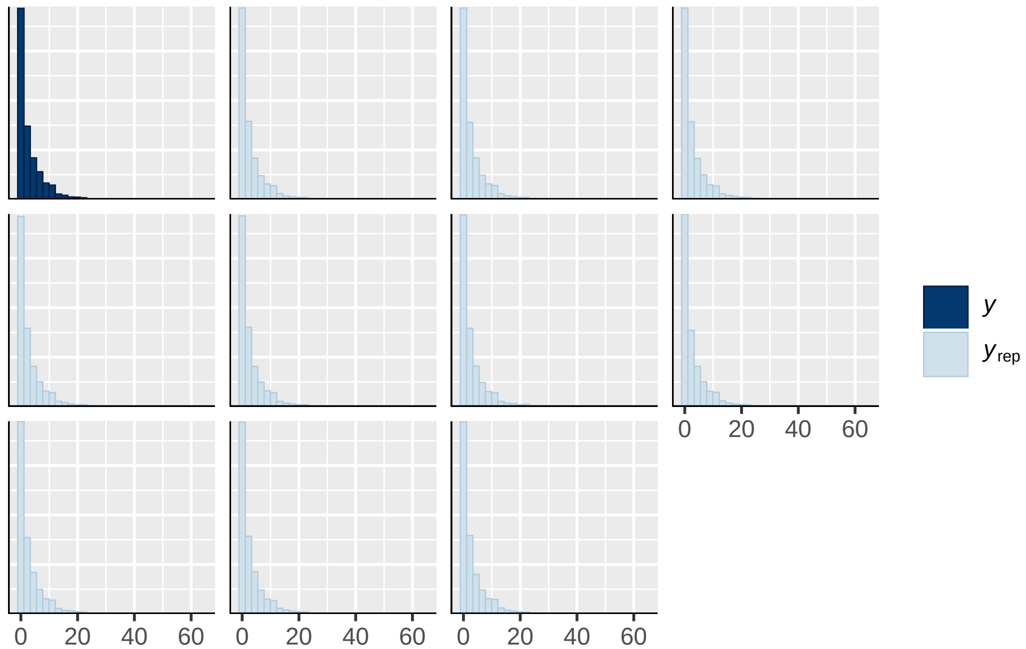
```
 student_t(3, 0, 2.5)        sd                                  Intercept
 student_t(3, 0, 2.5)        sd
 student_t(3, 0, 2.5)        sd                                  Intercept
 student_t(3, 0, 2.5)        sd
 student_t(3, 0, 2.5)        sd                                  Intercept
 student_t(3, 0, 2.5)        sd
 student_t(3, 0, 2.5)        sd                                  Intercept
        group resp dpar nlpar lb ub tag       source
                                                user
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                             default
                              0              default
    family_id                 0         (vectorized)
    family_id                 0         (vectorized)
       obs_id                 0         (vectorized)
       obs_id                 0         (vectorized)
participant_id                0         (vectorized)
participant_id                0         (vectorized)
      site_id                 0         (vectorized)
      site_id                 0         (vectorized)
```


b_Intercept
b_breast_development_p_sc
b_adhd_status_lifeY
b_age_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_inr_sc
breast_development_p_sc:adhd_status_lif
sd_family_id__Intercept
Chain
1
2
0 1000 2000 3000 4000

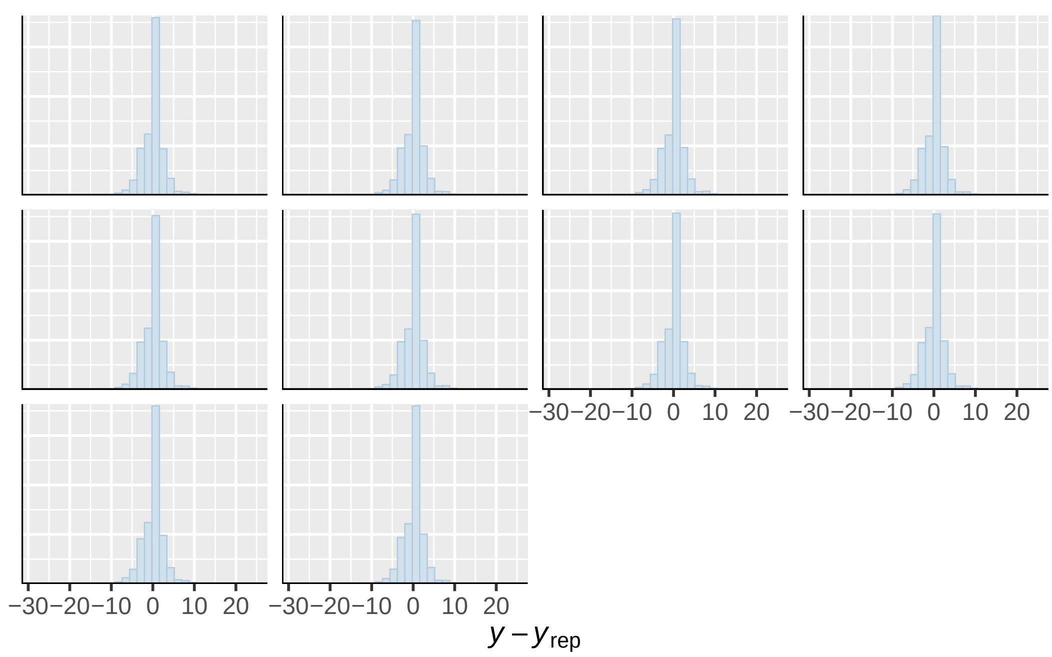




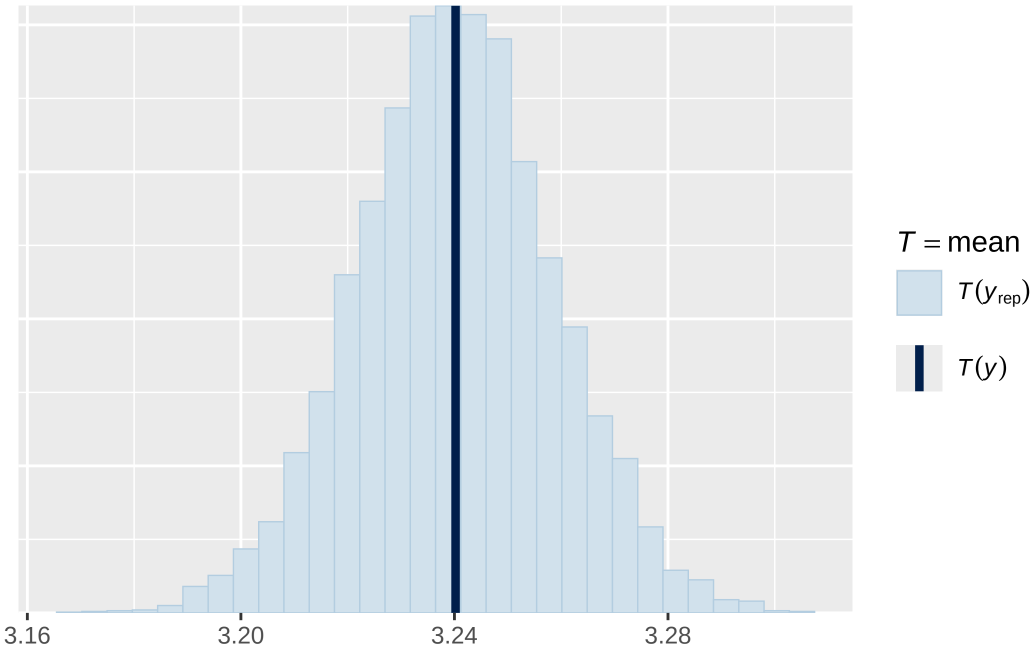

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



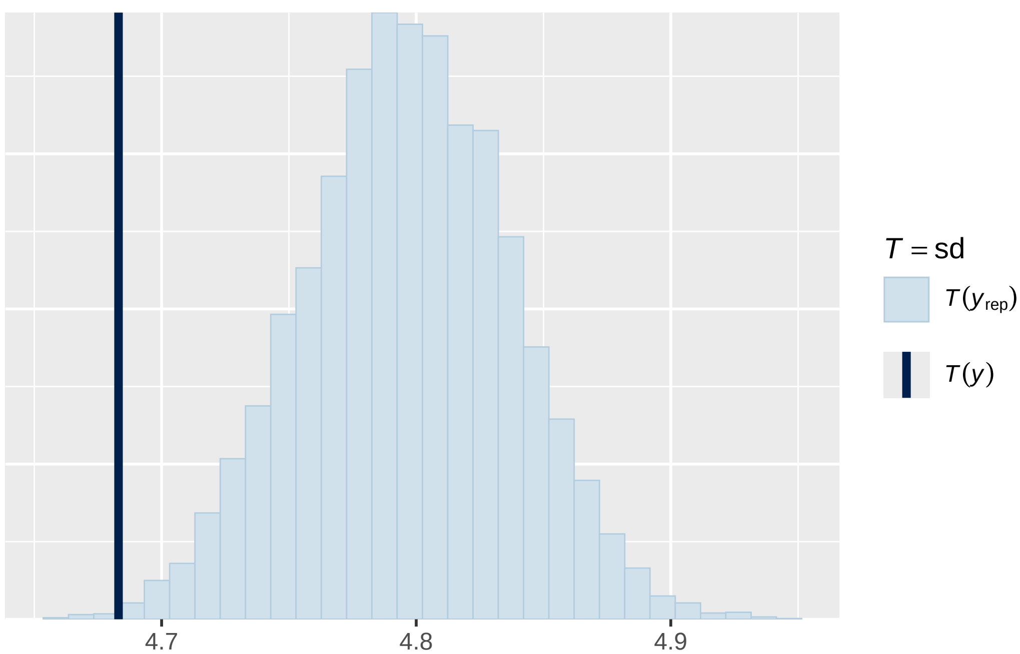

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

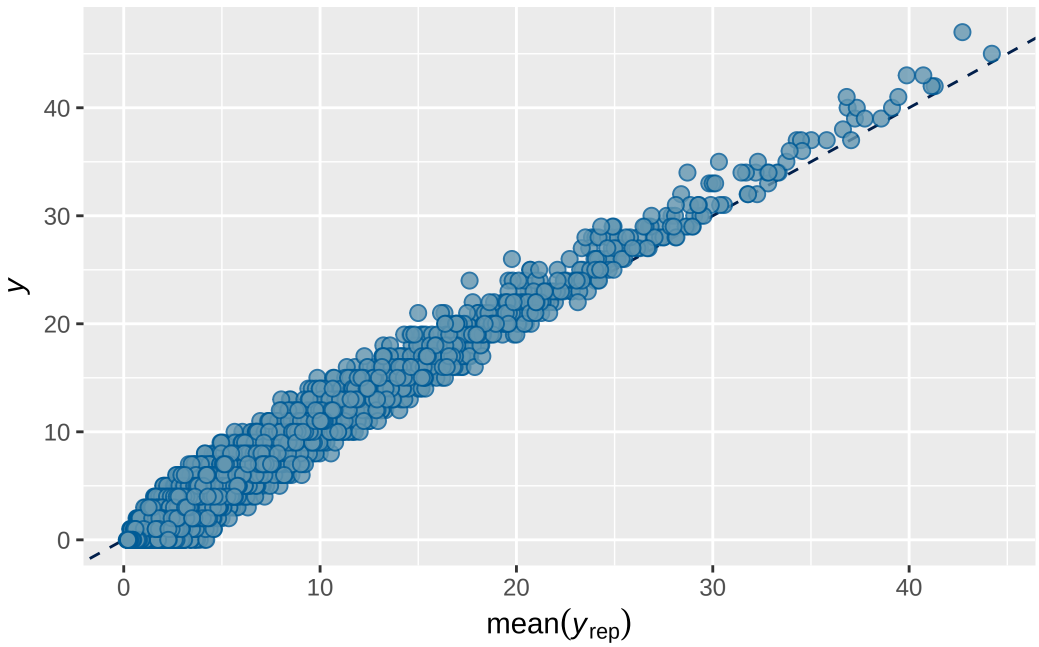



```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```





Model 4 output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.841   0.00237 0.836 0.845


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (1
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.648     0.018    0.612    0.683 1.002      797     1729

~obs_id (Number of levels: 19381)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.367     0.006    0.354    0.379 1.000     2687     4390

~participant_id (Number of levels: 5297)
```

```
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.462     0.020    0.423    0.501 1.005      585     1330

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.092     0.021    0.056    0.139 1.000     2747     4282

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            1.200     0.028    1.144    1.254 1.000
menarche_status_pY                   0.071     0.016    0.039    0.103 1.001
adhd_traits_sc                       0.815     0.016    0.785    0.846 1.001
age_sc                               0.057     0.014    0.030    0.084 1.000
ethnraceblack_nh                    -0.438     0.040   -0.515   -0.360 1.001
ethnracehispanic                    -0.080     0.037   -0.152   -0.007 1.000
ethnraceother                       -0.046     0.042   -0.130    0.034 1.001
inr_sc                              -0.023     0.018   -0.059    0.013 1.000
menarche_status_pY:adhd_traits_sc    0.041     0.020    0.001    0.081 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             3147     3872
menarche_status_pY                    5428     5483
adhd_traits_sc                        4505     4806
age_sc                                5473     5582
ethnraceblack_nh                      4531     5349
ethnracehispanic                      3486     4330
ethnraceother                         4434     4942
inr_sc                                5158     5553
menarche_status_pY:adhd_traits_sc     5025     5329

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 4 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 3 x 4
  Variable                          Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                <dbl>          <dbl>          <dbl>
1 Menarche Y [ADHD traits = mean]       7.38           4.02           10.9
2 ADHD traits [Menarche = 0]           15.6           14.9            16.2
3 Menarche × ADHD traits                0.74           0.02            1.45
```

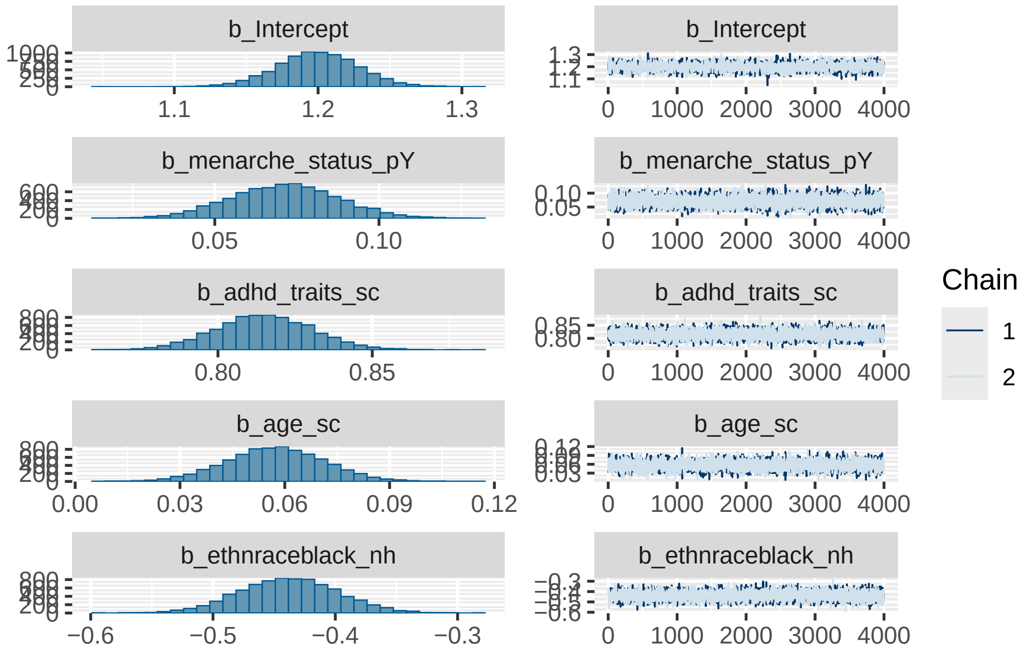
Model 4 interaction analysis

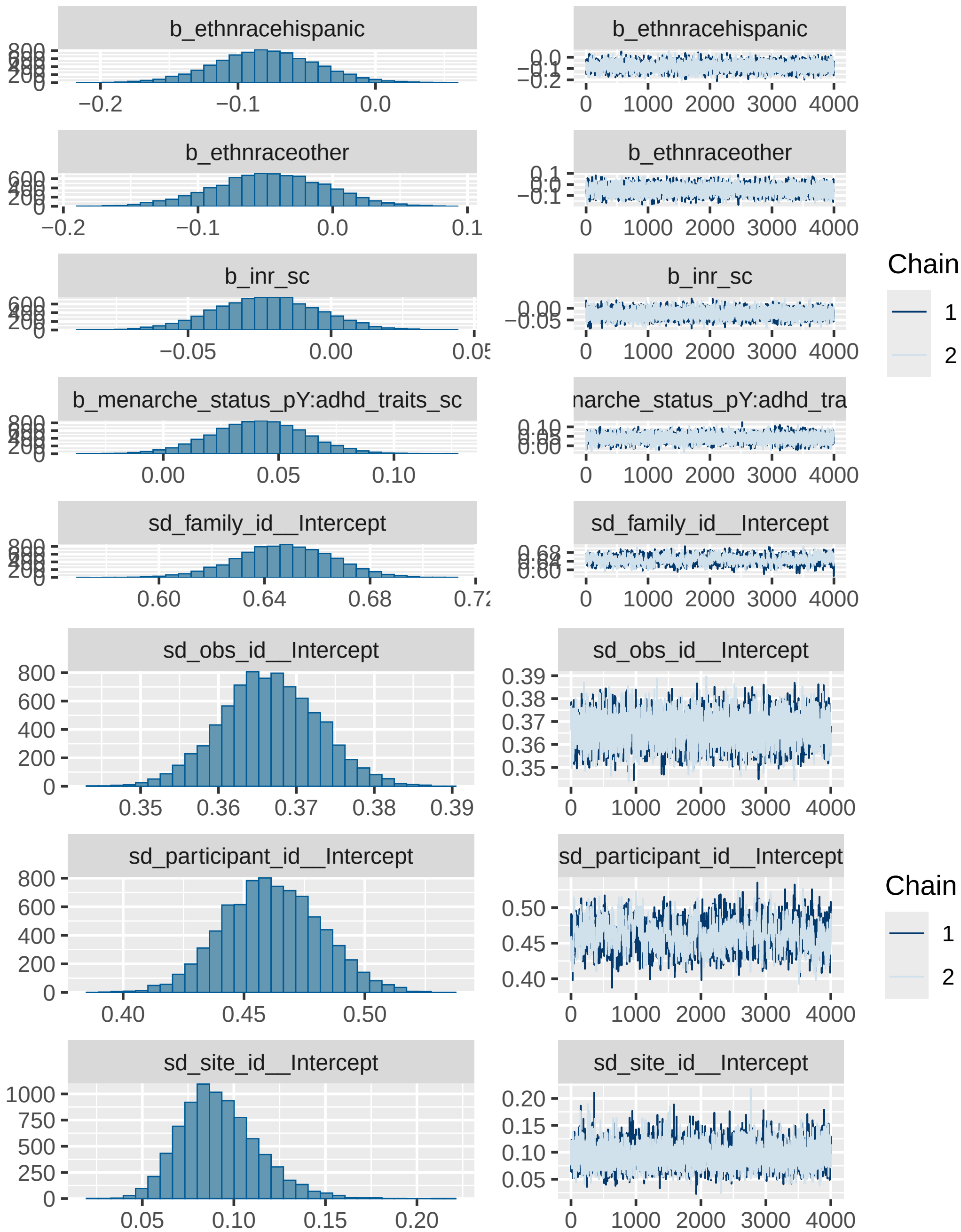
```
  menarche_status_p Estimate Lower 95% CI Upper 95% CI
1                 N    15.56        14.91        16.16
2                 Y    16.41        15.64        17.24
```

Model 4 diagnostics

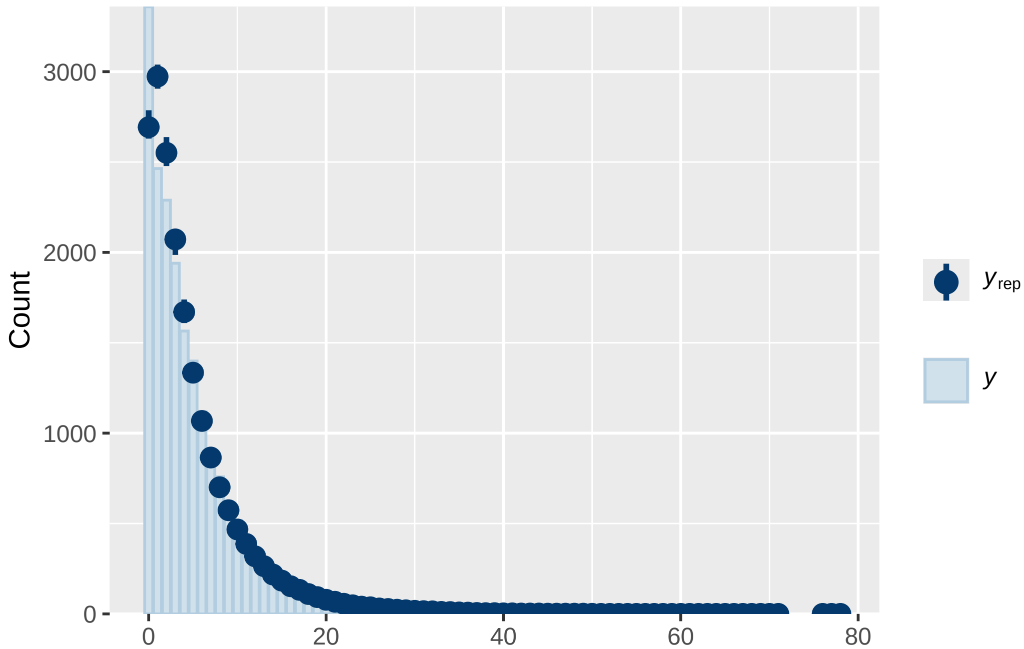
```
                prior     class                             coef
          normal(0, 1)         b
          normal(0, 1)         b                   adhd_traits_sc
          normal(0, 1)         b                           age_sc
          normal(0, 1)         b                 ethnraceblack_nh
          normal(0, 1)         b                 ethnracehispanic
          normal(0, 1)         b                    ethnraceother
          normal(0, 1)         b                           inr_sc
          normal(0, 1)         b               menarche_status_pY
          normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                        Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                        Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                        Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                        Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                               0              default
     family_id                 0         (vectorized)
     family_id                 0         (vectorized)
```

```
        obs_id                  0         (vectorized)
        obs_id                  0         (vectorized)
participant_id                  0         (vectorized)
participant_id                  0         (vectorized)
       site_id                  0         (vectorized)
       site_id                  0         (vectorized)
```

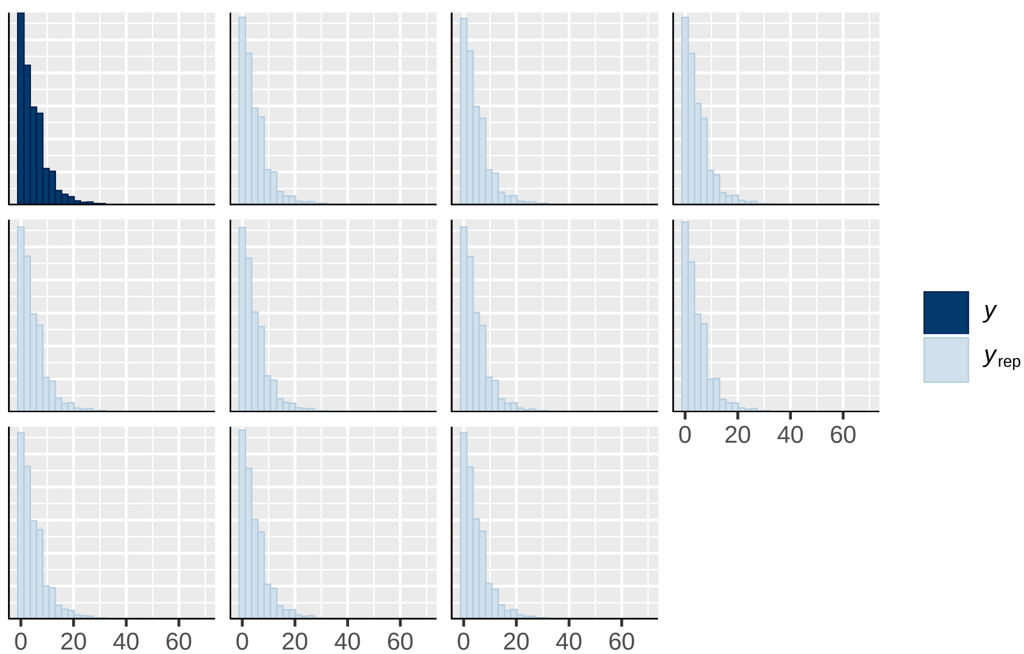




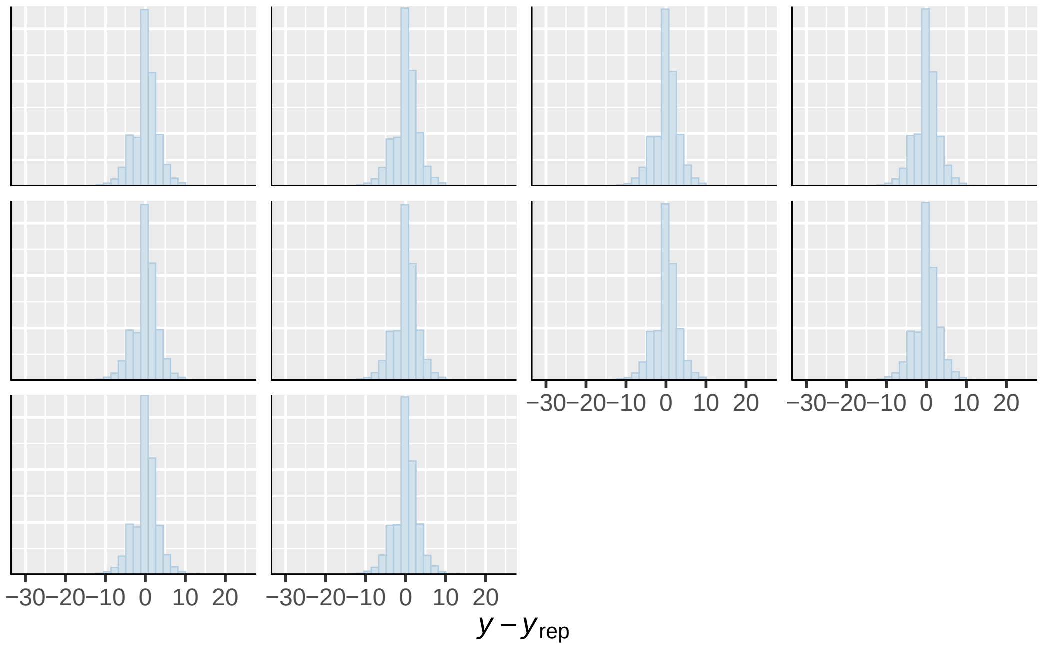
b_ethnracehispanic
−0.2
−0.1
0.0
b_ethnracehispanic
0
1000
2000
3000
4000
b_ethnraceother
−0.2
−0.1
0.0
0.1
b_ethnraceother
0
1000
2000
3000
4000
Chain
1
2
b_inr_sc
−0.05
0.00
b_inr_sc
0.00
−0.05
0
1000
2000
3000
4000
b_menarche_status_pY:adhd_traits_sc
0.00
0.05
0.10
narche_status_pY:adhd_tra
0
1000
2000
3000
4000
sd_family_id__Intercept
0.60
0.64
0.68
sd_family_id__Intercept
0
1000
2000
3000
4000
sd_obs_id__Intercept
800
600
400
200
0
0.35
0.36
0.37
0.38
0.39
sd_obs_id__Intercept
0.39
0.38
0.37
0.36
0.35
0
1000
2000
3000
4000
sd_participant_id__Intercept
800
600
400
200
0
0.40
0.45
0.50
sd_participant_id__Intercept
0.50
0.45
0.40
0
1000
2000
3000
4000
Chain
1
2
sd_site_id__Intercept
1000
750
500
250
0
0.05
0.10
0.15
0.20
sd_site_id__Intercept
0.20
0.15
0.10
0.05
0
1000
2000
3000
4000



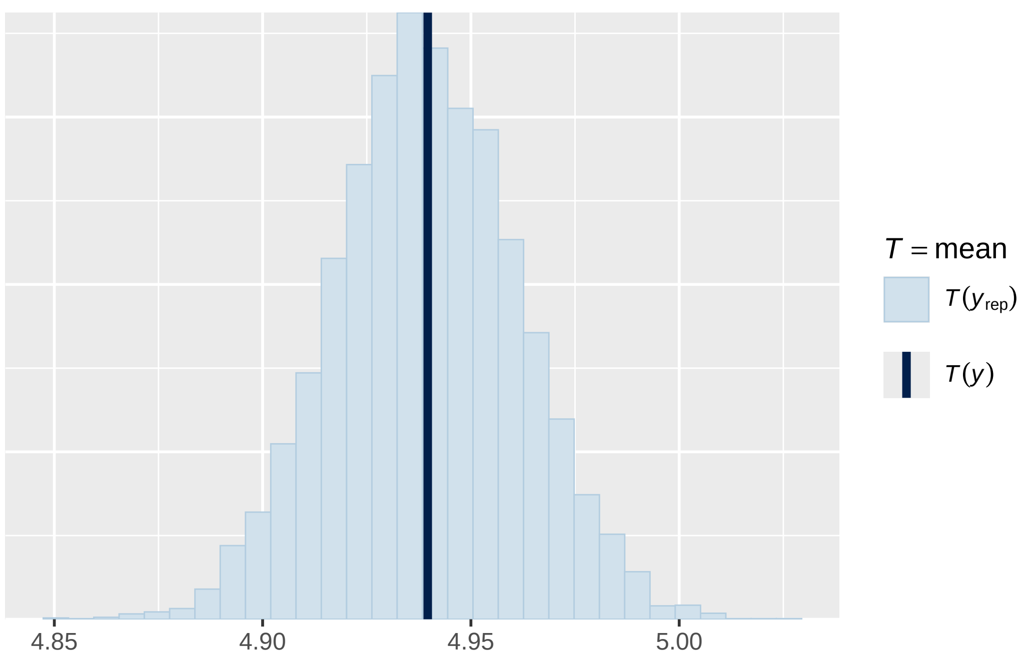
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



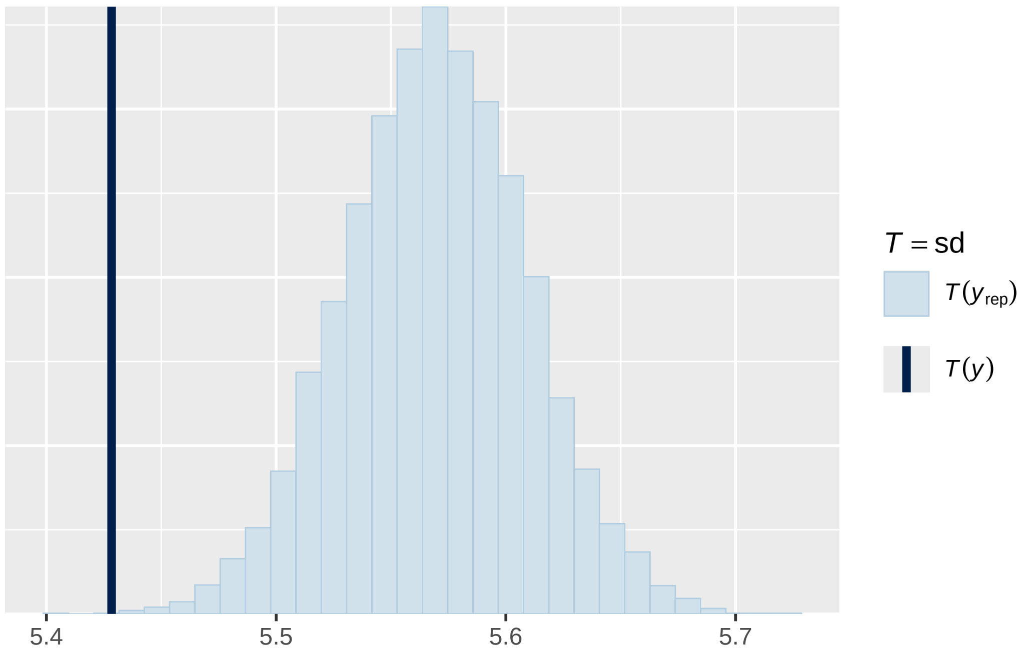
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

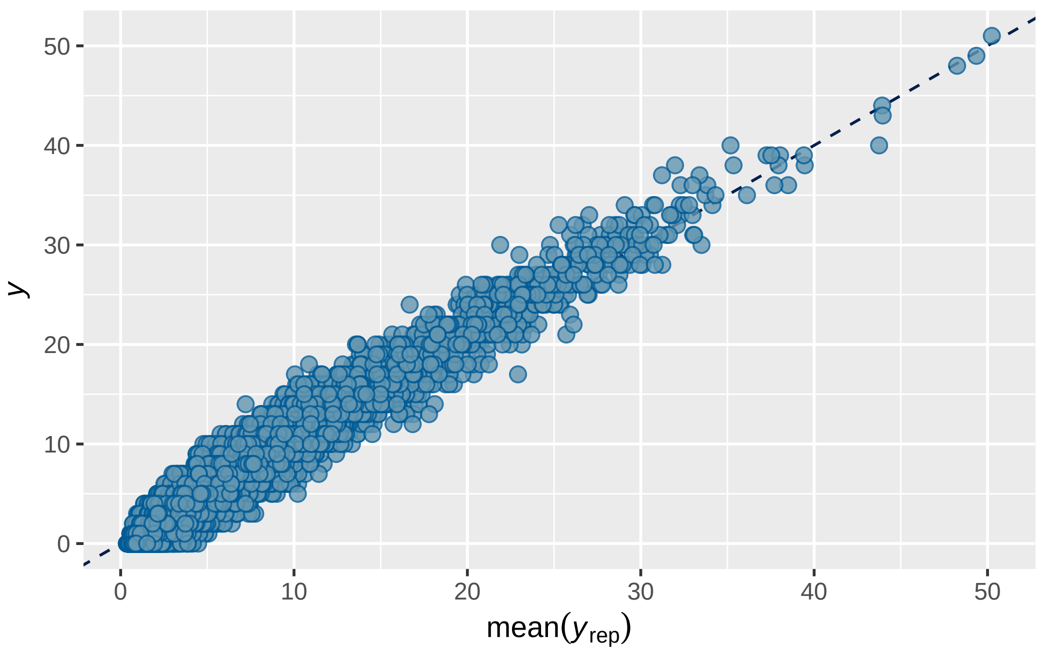


`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



`stat_bin()` using `bins = 30`. Pick better value `binwidth`.


5.4
5.5
5.6
5.7
$T = \mathrm{sd}$
$T(y_{\mathrm{rep}})$
$T(y)$


50
40
30
20
10
0
y
0
10
20
30
40
50
mean(y_{rep})

Model 4b output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.841    0.0024 0.836 0.845


 Family: poisson
  Links: mu = log
Formula: internalising ~ breast_development_p_sc * adhd_traits_sc + age_sc + ethnrace + inr_s
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.649     0.018    0.614    0.683 1.000     1116     2247

~obs_id (Number of levels: 19364)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.366     0.006    0.354    0.378 1.000     3108     5114

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.461     0.020    0.423    0.500 1.000      850     1715

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.092     0.022    0.055    0.142 1.000     3971     4763

Regression Coefficients:
                                      Estimate Est.Error l-95% CI u-95% CI
Intercept                                1.221     0.028    1.162    1.274
breast_development_p_sc                  0.057     0.017    0.025    0.089
adhd_traits_sc                           0.827     0.014    0.799    0.854
age_sc                                   0.069     0.014    0.042    0.096
ethnraceblack_nh                        -0.431     0.040   -0.509   -0.353
ethnracehispanic                        -0.074     0.036   -0.146   -0.003
ethnraceother                           -0.046     0.040   -0.126    0.033
inr_sc                                  -0.019     0.019   -0.056    0.018
breast_development_p_sc:adhd_traits_sc   0.021     0.021   -0.020    0.063
                                       Rhat Bulk_ESS Tail_ESS
Intercept                             1.001     4055     4623
```

```
breast_development_p_sc                1.000     6862     6647
adhd_traits_sc                         1.000     6662     5946
age_sc                                 1.000     6712     5885
ethnraceblack_nh                       1.000     5460     5714
ethnracehispanic                       1.001     4816     4652
ethnraceother                          1.001     4712     5239
inr_sc                                 1.000     7281     5925
breast_development_p_sc:adhd_traits_sc 1.000     7014     6048

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

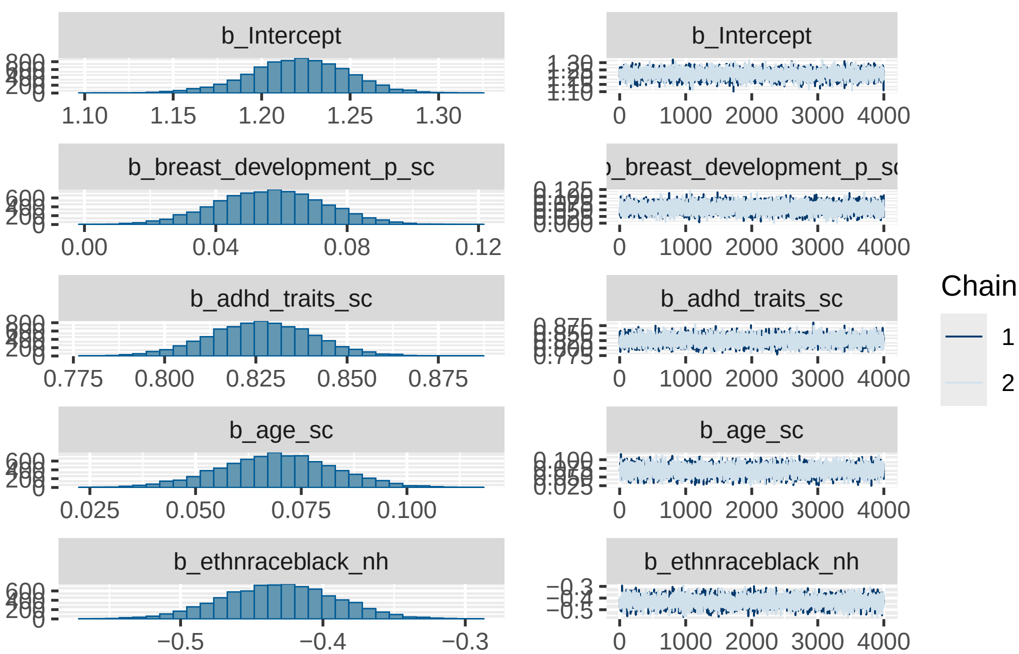
Model 4b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

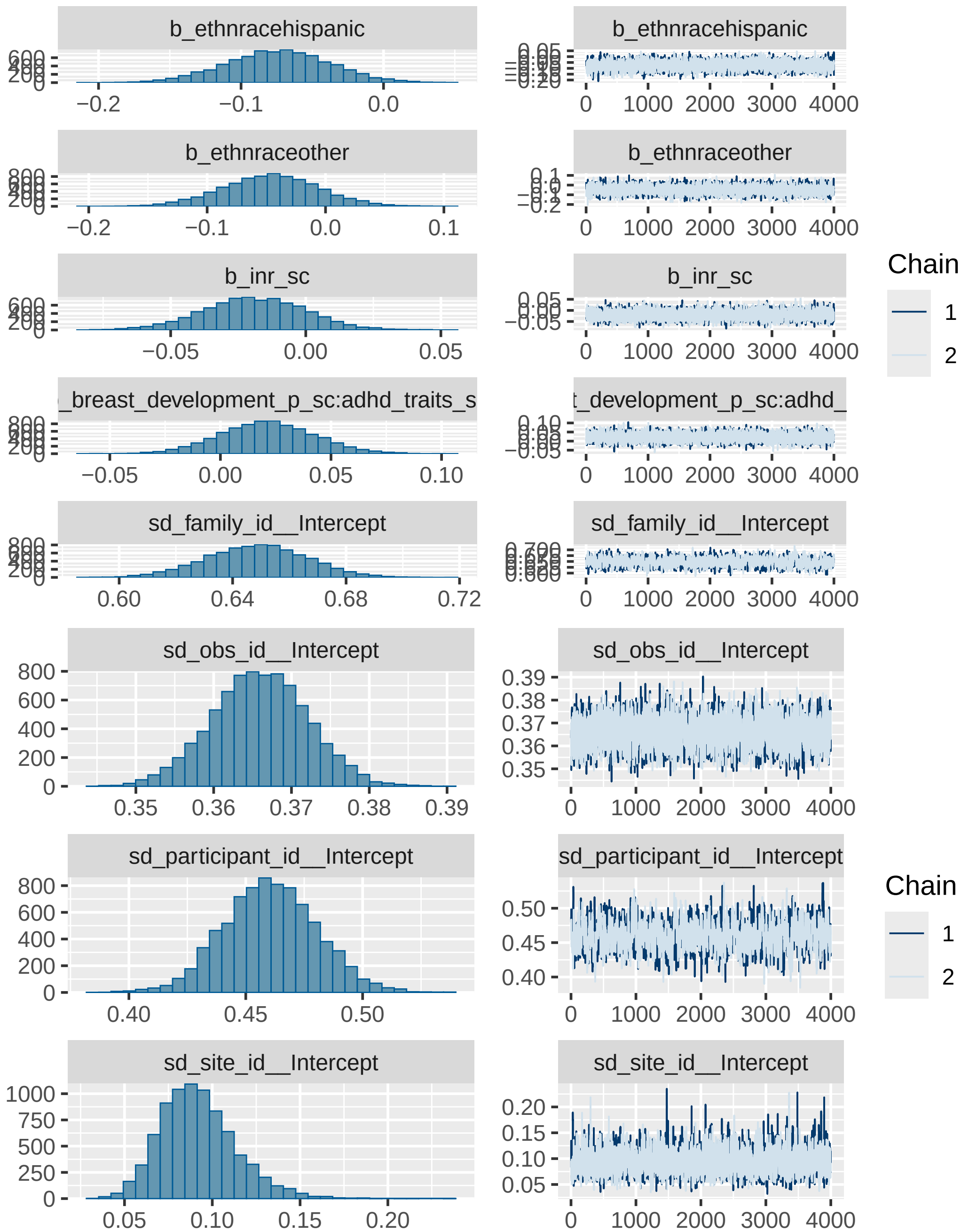
```
# A tibble: 3 x 4
  Variable                                 Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                       <dbl>          <dbl>          <dbl>
1 Breast development (per 1-unit) [ADHD ~     3.55           1.55           5.65
2 ADHD traits (per 1-unit) [Breast devel~    15.8           15.2           16.4
3 Interaction (Breast development × ADHD~     0.23          -0.22           0.69
```

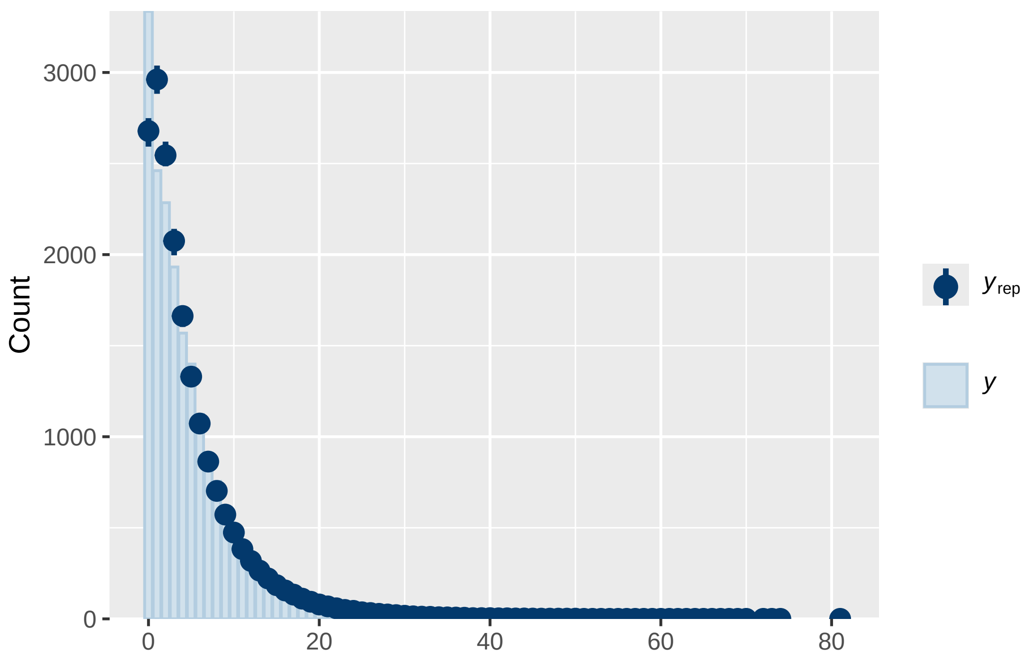
Model 4b diagnostics

```
                prior     class                                   coef
         normal(0, 1)         b
         normal(0, 1)         b                         adhd_traits_sc
         normal(0, 1)         b                                 age_sc
         normal(0, 1)         b                breast_development_p_sc
         normal(0, 1)         b breast_development_p_sc:adhd_traits_sc
         normal(0, 1)         b                       ethnraceblack_nh
         normal(0, 1)         b                       ethnracehispanic
         normal(0, 1)         b                          ethnraceother
         normal(0, 1)         b                                 inr_sc
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                              Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                              Intercept
```

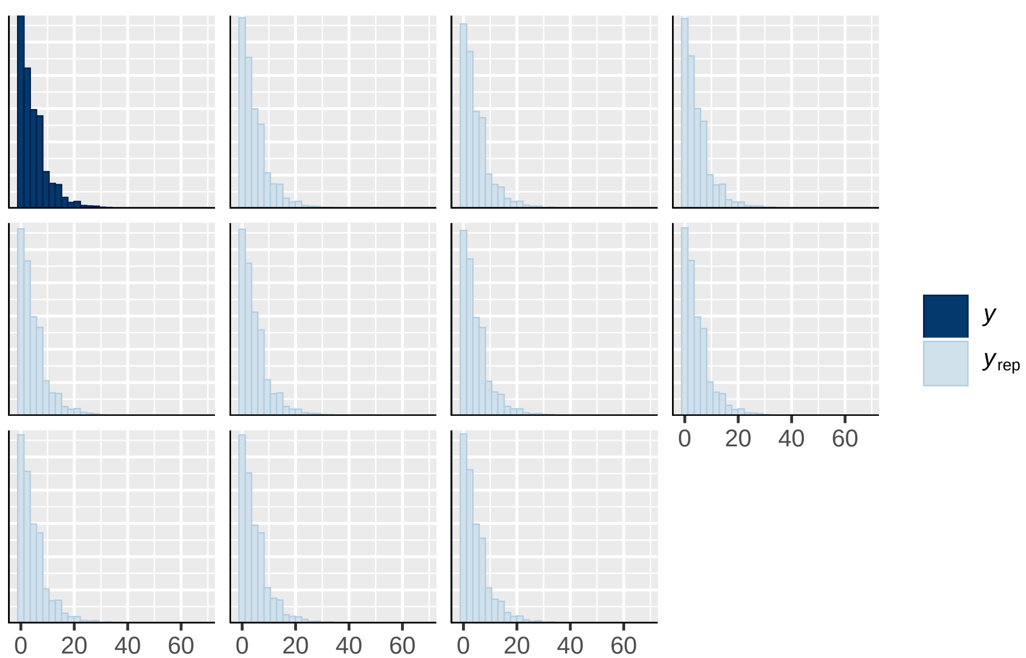
```
student_t(3, 0, 2.5)        sd                                  
student_t(3, 0, 2.5)        sd                                      Intercept
student_t(3, 0, 2.5)        sd                                  
student_t(3, 0, 2.5)        sd                                      Intercept
       group resp dpar nlpar lb ub tag        source
                                                user
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                             default
                              0              default
   family_id                  0         (vectorized)
   family_id                  0         (vectorized)
      obs_id                  0         (vectorized)
      obs_id                  0         (vectorized)
participant_id                0         (vectorized)
participant_id                0         (vectorized)
     site_id                  0         (vectorized)
     site_id                  0         (vectorized)
```



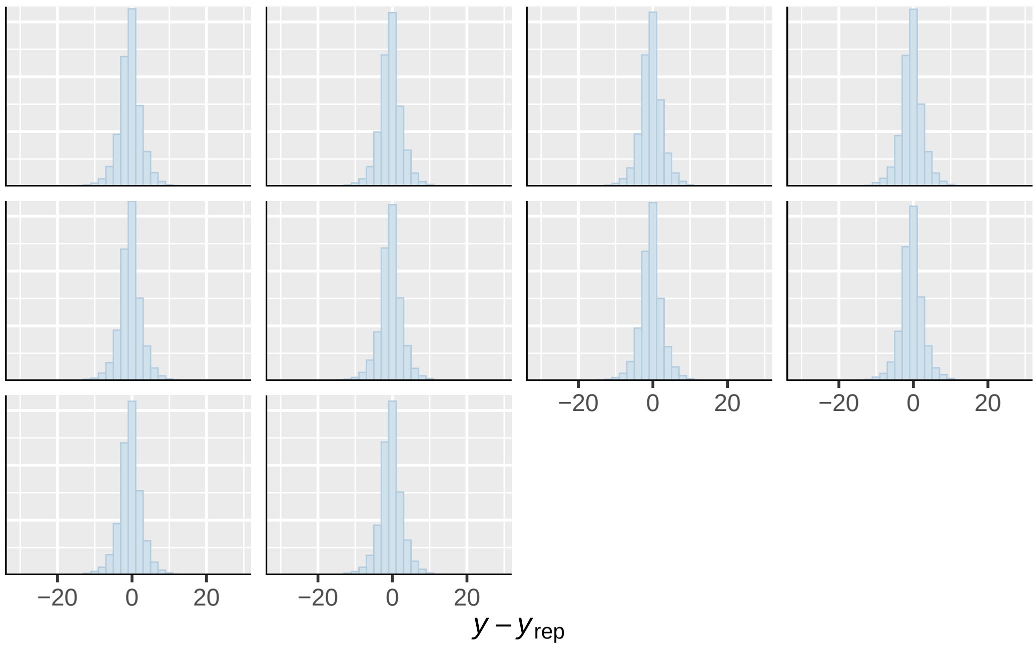

b_ethnracehispanic
b_ethnraceother
b_inr_sc
_breast_development_p_sc:adhd_traits_s
t_development_p_sc:adhd_
sd_family_id__Intercept
sd_obs_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2



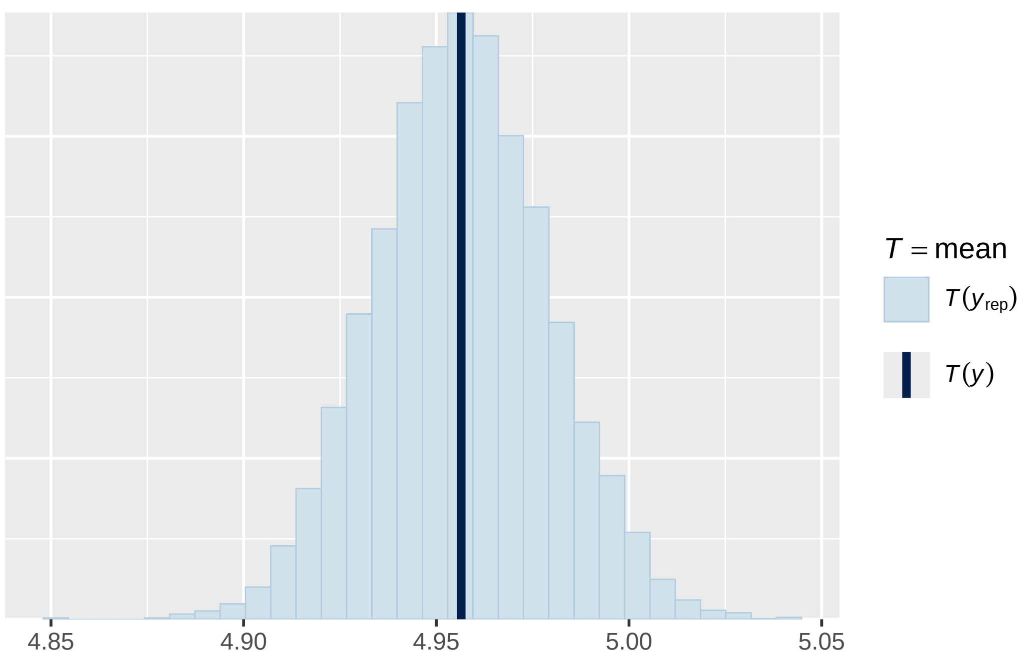

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



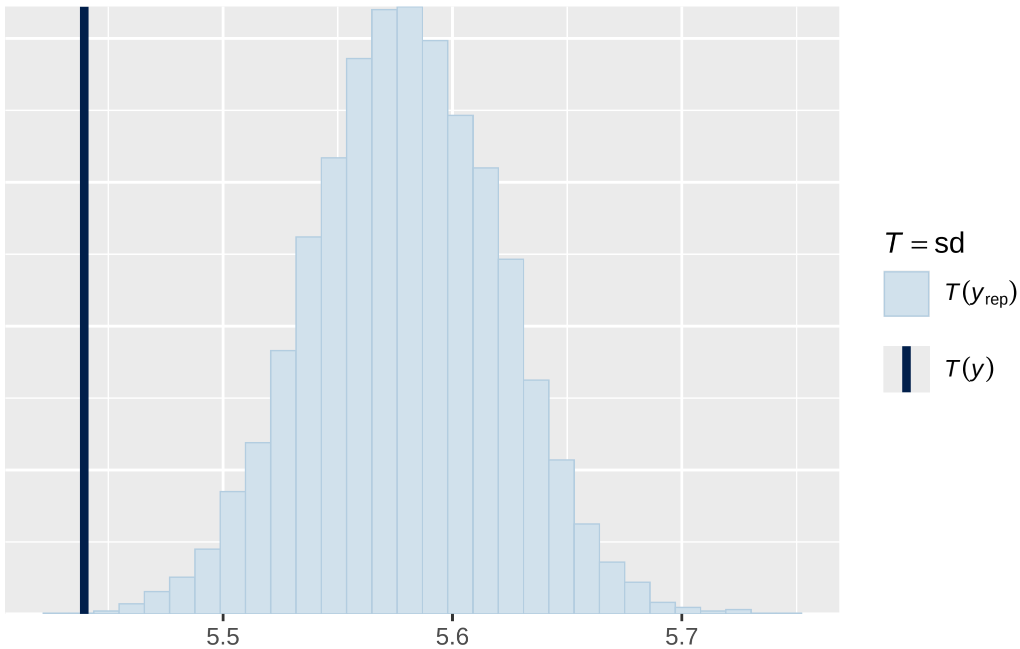

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

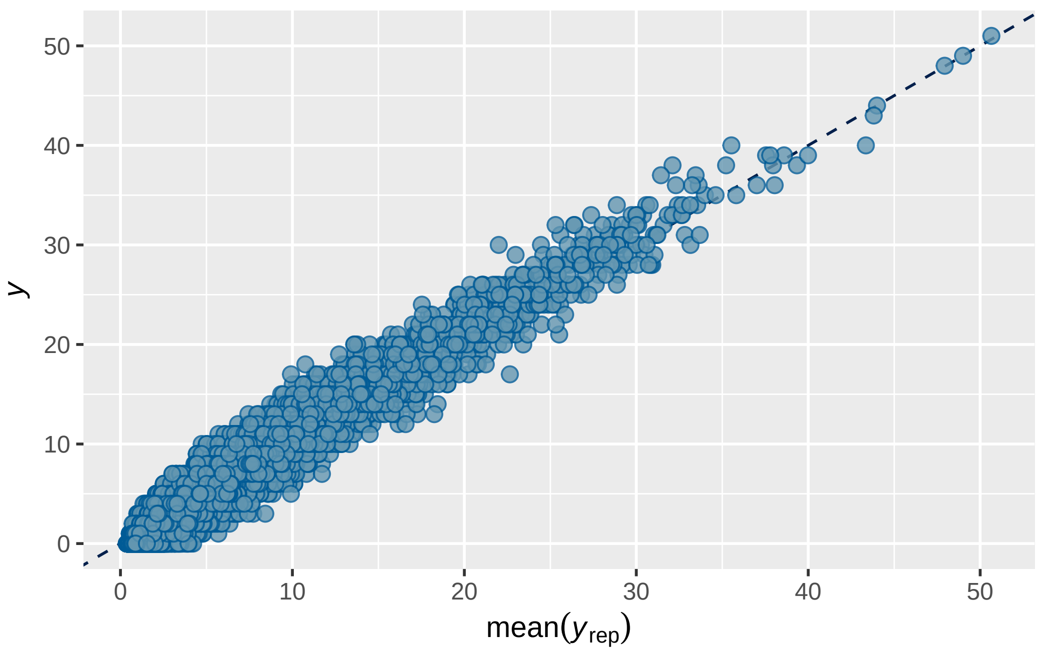


`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



`stat_bin()` using `bins = 30`. Pick better value `binwidth`.





Model 5 output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.863   0.00233 0.858 0.867
```

```
 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.719     0.031    0.656    0.776 1.013      465      919

~obs_id (Number of levels: 19381)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.373     0.008    0.357    0.389 1.000     1721     2867

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.724     0.028    0.673    0.781 1.013      486      826

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.111     0.027    0.065    0.172 1.000     3767     3204

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.412     0.035    0.341    0.480 1.001
menarche_status_pY                   0.079     0.020    0.039    0.118 1.001
adhd_traits_sc                       1.077     0.019    1.040    1.114 1.000
age_sc                              -0.123     0.017   -0.157   -0.089 1.000
ethnraceblack_nh                    -0.067     0.052   -0.170    0.037 1.001
ethnracehispanic                    -0.044     0.048   -0.138    0.048 1.001
ethnraceother                       -0.052     0.053   -0.155    0.050 1.001
inr_sc                              -0.152     0.024   -0.197   -0.103 1.000
menarche_status_pY:adhd_traits_sc    0.048     0.023    0.003    0.095 1.001
                                  Bulk_ESS Tail_ESS
Intercept                             4048     2842
menarche_status_pY                    5891     3346
adhd_traits_sc                        5134     3151
age_sc                                6012     3444
ethnraceblack_nh                      5231     2990
ethnracehispanic                      4835     3020
ethnraceother                         5475     3266
```

```
inr_sc                               5788     2679
menarche_status_pY:adhd_traits_sc    5362     3285

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 5 effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 3 x 4
  Variable                        Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                              <dbl>          <dbl>          <dbl>
1 Menarche Y [ADHD traits = mean]     8.17           3.95           12.6
2 ADHD traits [Menarche = 0]         21.1           20.3            21.9
3 Menarche × ADHD traits              0.86           0.06            1.7
```

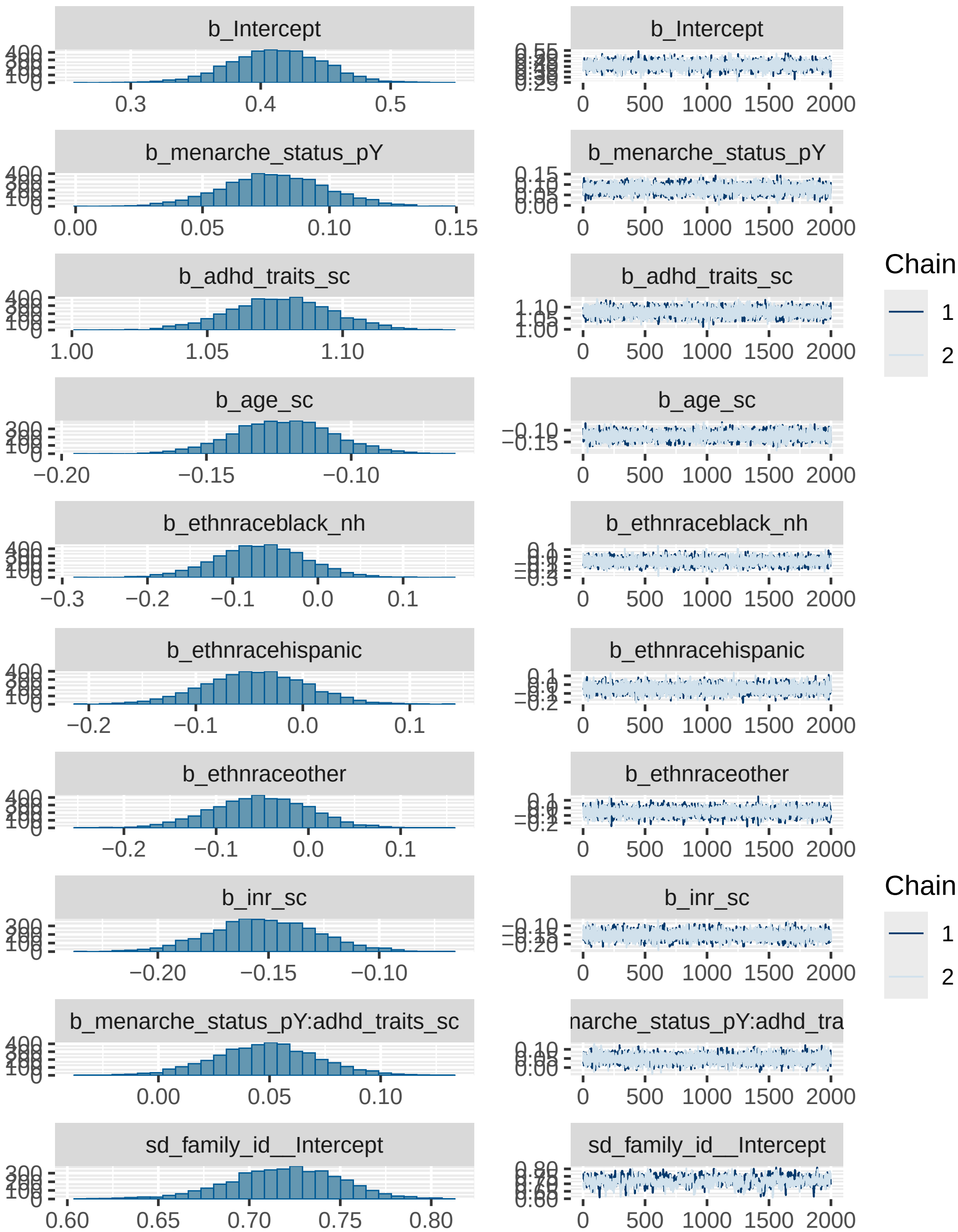
Model 5 interaction analysis

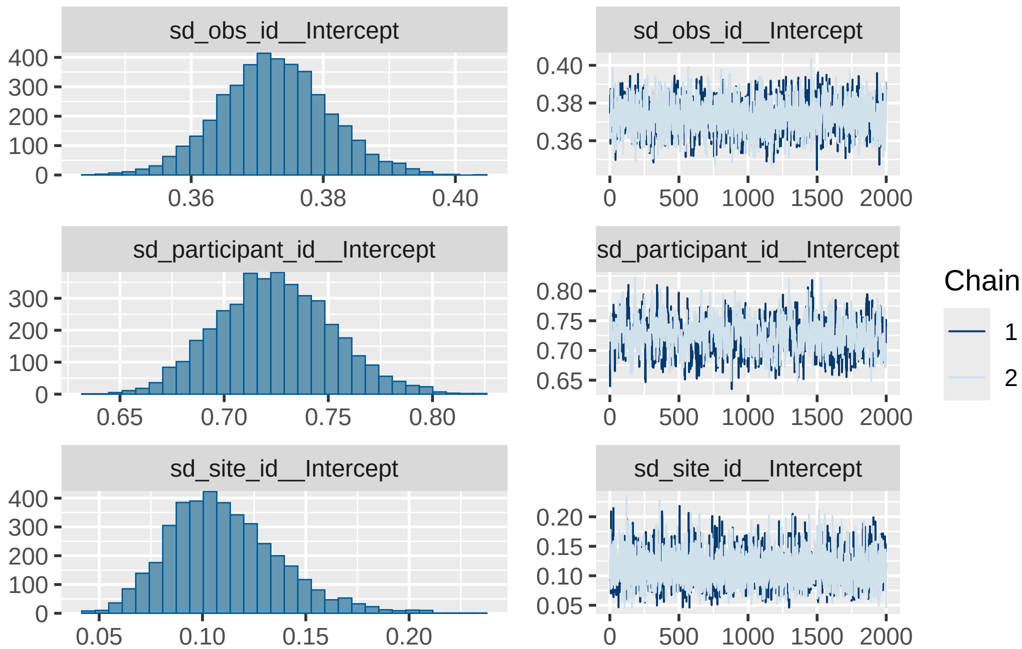
```
  menarche_status_p Estimate Lower 95% CI Upper 95% CI
1                 N    21.07        20.23        21.81
2                 Y    22.12        21.17        23.17
```

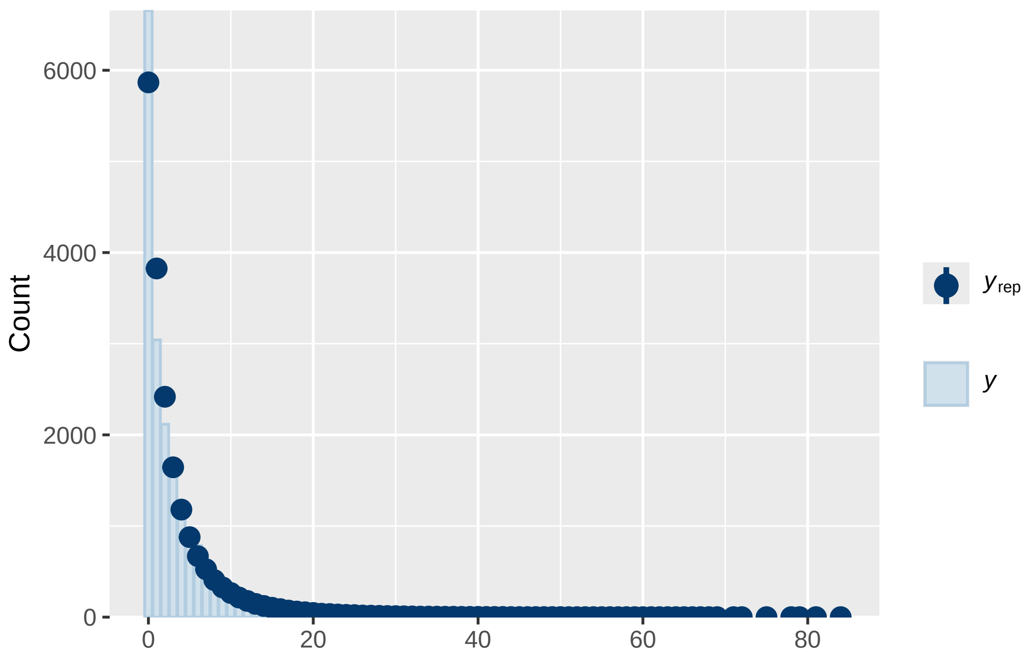
Model 5 diagnostics

```
              prior     class                            coef
        normal(0, 1)         b
        normal(0, 1)         b                  adhd_traits_sc
        normal(0, 1)         b                          age_sc
        normal(0, 1)         b               ethnraceblack_nh
        normal(0, 1)         b               ethnracehispanic
        normal(0, 1)         b                  ethnraceother
        normal(0, 1)         b                          inr_sc
        normal(0, 1)         b              menarche_status_pY
        normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                       Intercept
 student_t(3, 0, 2.9)        sd
```

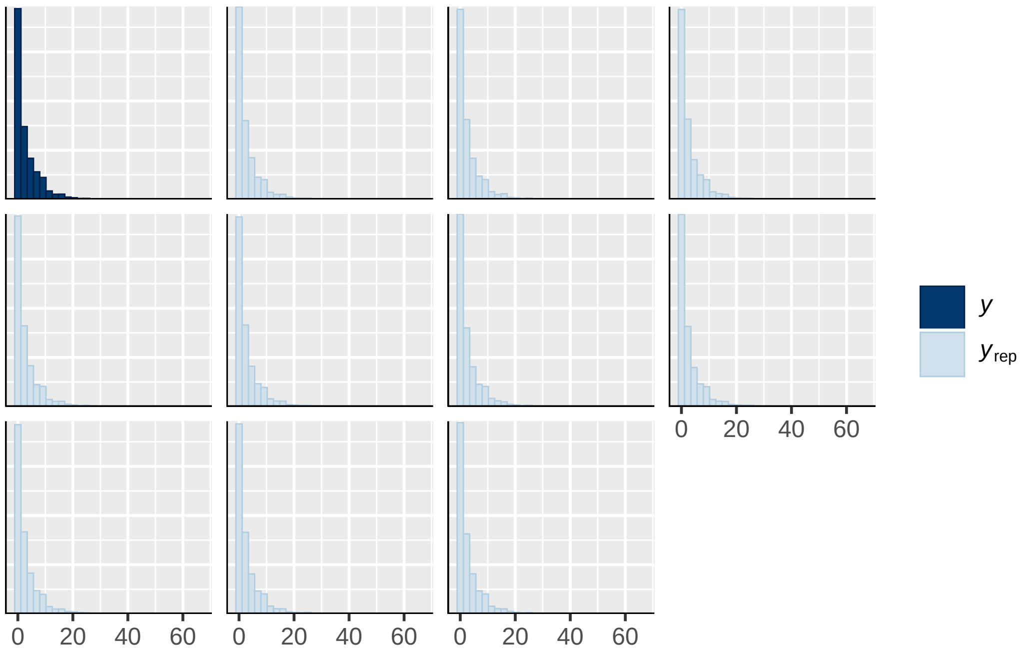
```
student_t(3, 0, 2.9)        sd                          Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                          Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                          Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
        obs_id                  0        (vectorized)
        obs_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
0.3
0.4
0.5
b_Intercept
0 500 1000 1500 2000
b_menarche_status_pY
0.00
0.05
0.10
0.15
b_menarche_status_pY
0.15
0 500 1000 1500 2000
b_adhd_traits_sc
1.00
1.05
1.10
b_adhd_traits_sc
1.10
1.05
1.00
0 500 1000 1500 2000
b_age_sc
−0.20
−0.15
−0.10
b_age_sc
−0.10
−0.15
0 500 1000 1500 2000
b_ethnraceblack_nh
−0.3
−0.2
−0.1
0.0
0.1
b_ethnraceblack_nh
0 500 1000 1500 2000
b_ethnracehispanic
−0.2
−0.1
0.0
0.1
b_ethnracehispanic
0 500 1000 1500 2000
b_ethnraceother
−0.2
−0.1
0.0
0.1
b_ethnraceother
0 500 1000 1500 2000
b_inr_sc
−0.20
−0.15
−0.10
b_inr_sc
0 500 1000 1500 2000
b_menarche_status_pY:adhd_traits_sc
0.00
0.05
0.10
narche_status_pY:adhd_tra
0 500 1000 1500 2000
sd_family_id__Intercept
0.60
0.65
0.70
0.75
0.80
sd_family_id__Intercept
0 500 1000 1500 2000
Chain
1
2
Chain
1
2

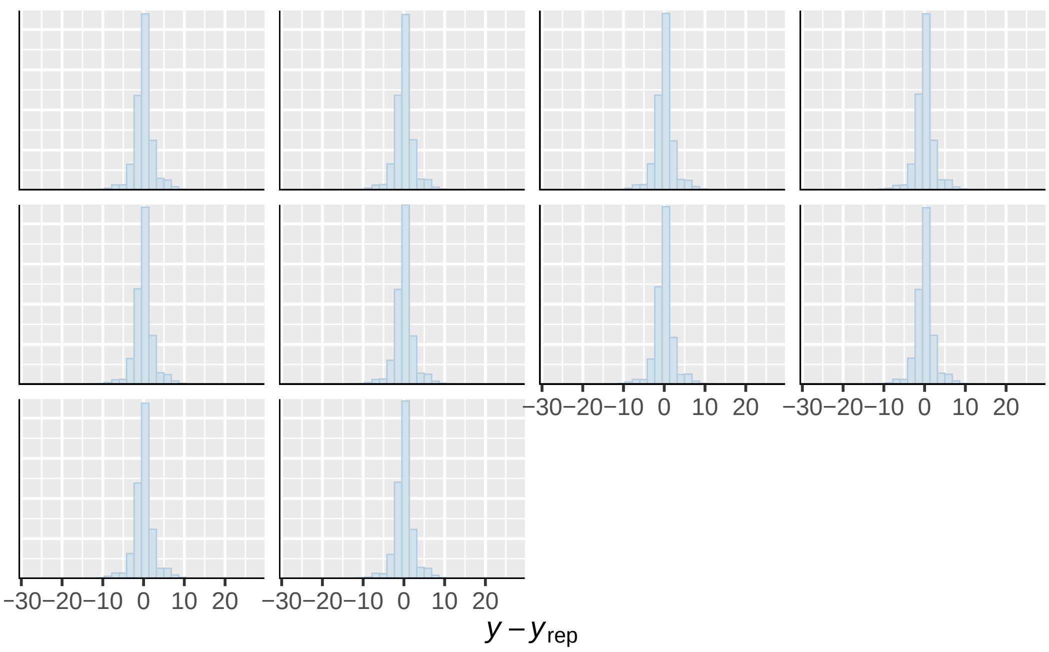




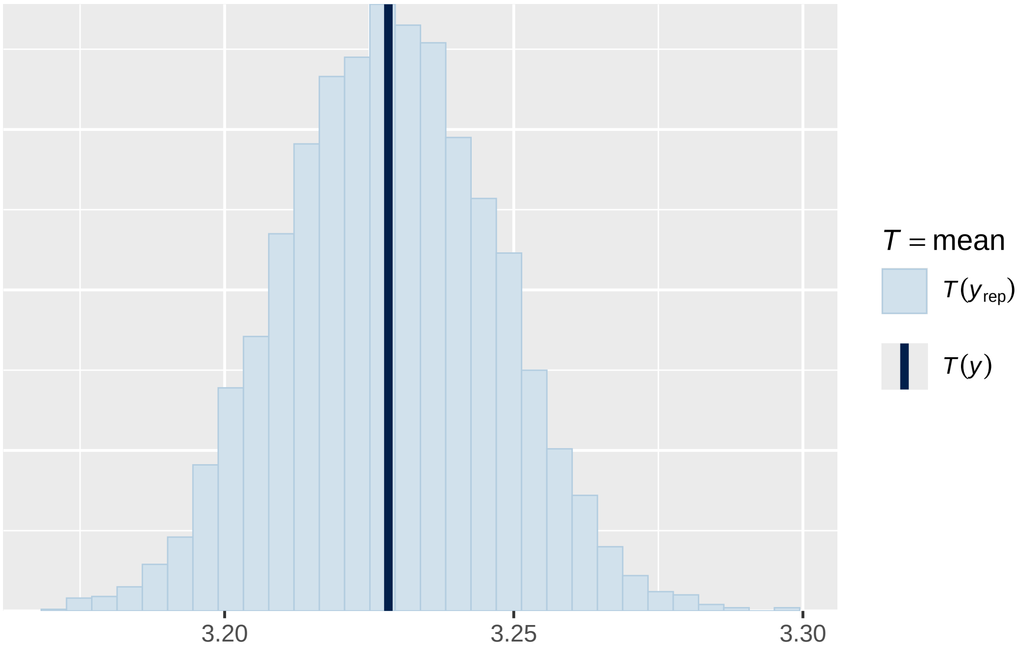

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



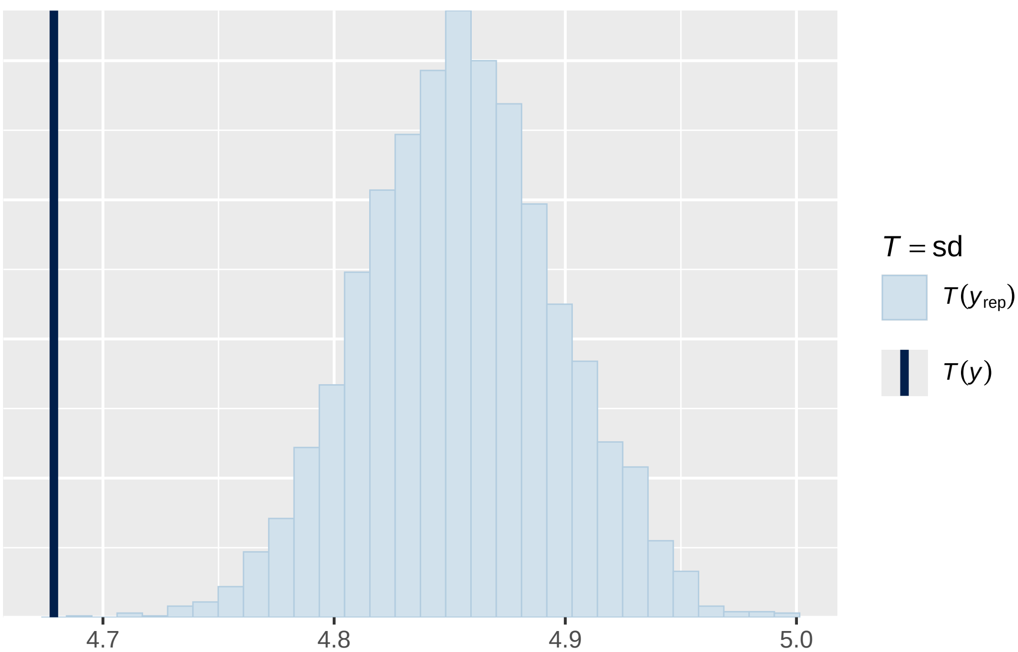
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

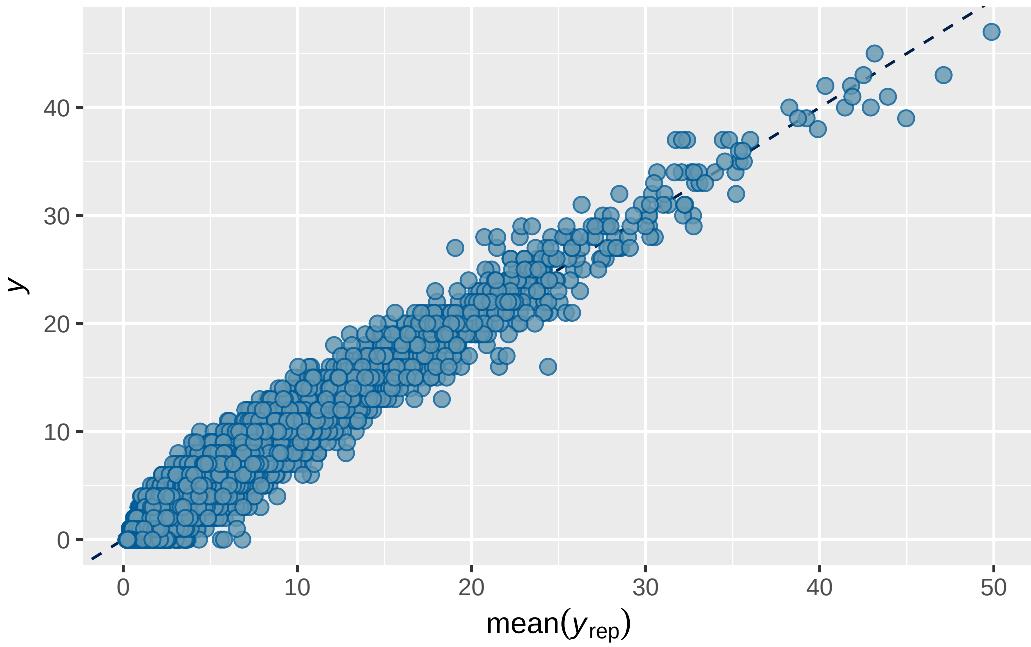


`stat_bin()` using `bins = 30`. Pick better value `binwidth`.




```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```





Model 5b output

```
   Estimate Est.Error  Q2.5 Q97.5
R2    0.863   0.00233 0.858 0.867


 Family: poisson
  Links: mu = log
Formula: externalising ~ breast_development_p_sc * adhd_traits_sc + age_sc + ethnrace + inr_s
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.714     0.029    0.653    0.767 1.001      631     1017

~obs_id (Number of levels: 19364)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.369     0.008    0.353    0.385 1.002     1510     2684

~participant_id (Number of levels: 5289)
```

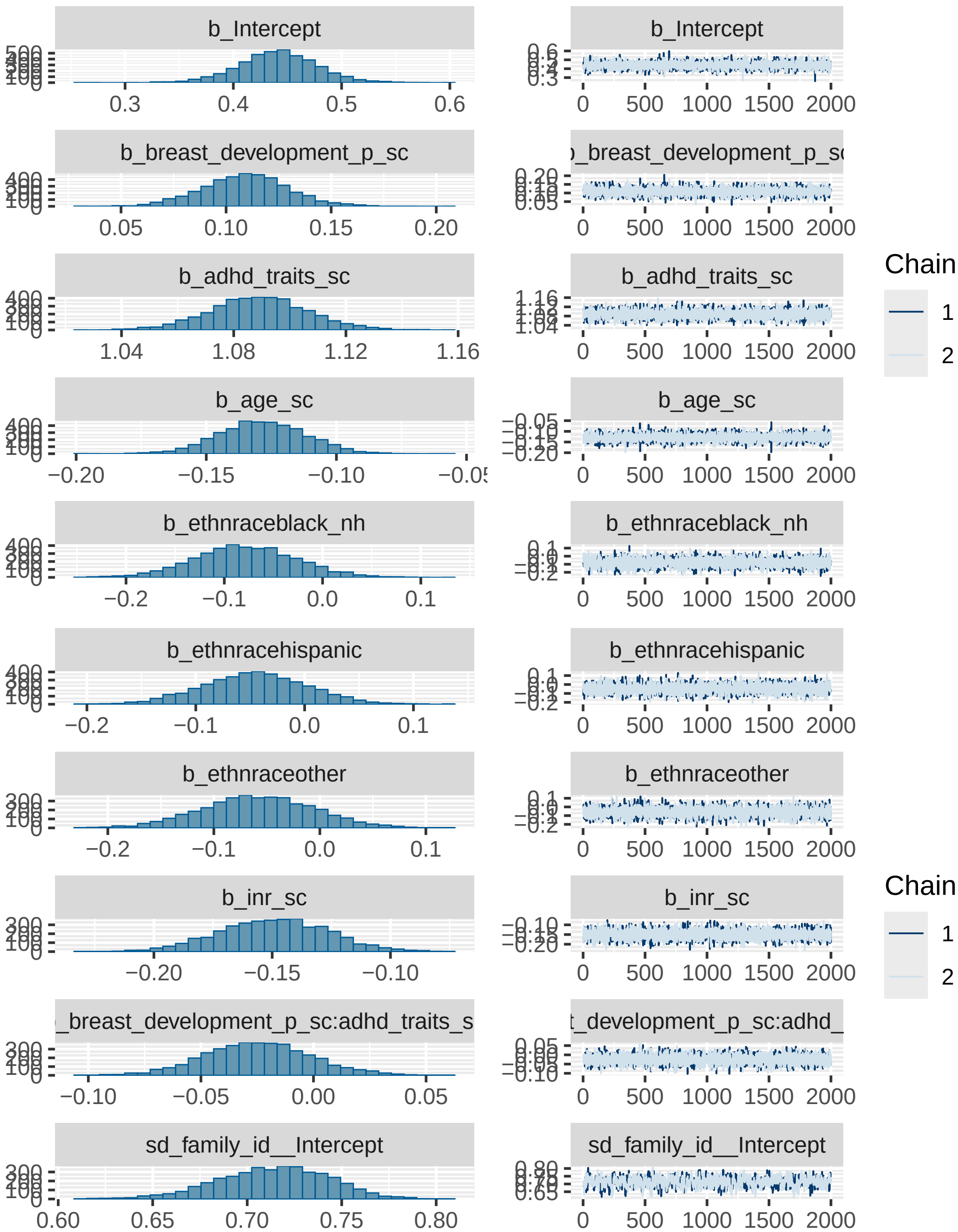
```
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.728     0.028    0.676    0.784 1.002      649      866

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.112     0.027    0.066    0.172 1.001     3311     2815

Regression Coefficients:
                                       Estimate Est.Error l-95% CI u-95% CI
Intercept                                 0.441     0.036    0.369    0.512
breast_development_p_sc                   0.109     0.020    0.069    0.150
adhd_traits_sc                            1.089     0.017    1.056    1.124
age_sc                                   -0.128     0.017   -0.160   -0.097
ethnraceblack_nh                         -0.078     0.052   -0.178    0.025
ethnracehispanic                         -0.045     0.048   -0.138    0.047
ethnraceother                            -0.058     0.053   -0.161    0.044
inr_sc                                   -0.149     0.023   -0.194   -0.104
breast_development_p_sc:adhd_traits_sc   -0.025     0.024   -0.070    0.023
                                        Rhat Bulk_ESS Tail_ESS
Intercept                              1.000     4227     2777
breast_development_p_sc                1.001     6233     3262
adhd_traits_sc                         1.000     5412     3079
age_sc                                 1.003     5752     3257
ethnraceblack_nh                       1.001     5533     2933
ethnracehispanic                       1.001     4499     3261
ethnraceother                          1.000     5564     2852
inr_sc                                 1.000     7177     3194
breast_development_p_sc:adhd_traits_sc 1.001     6426     3452

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

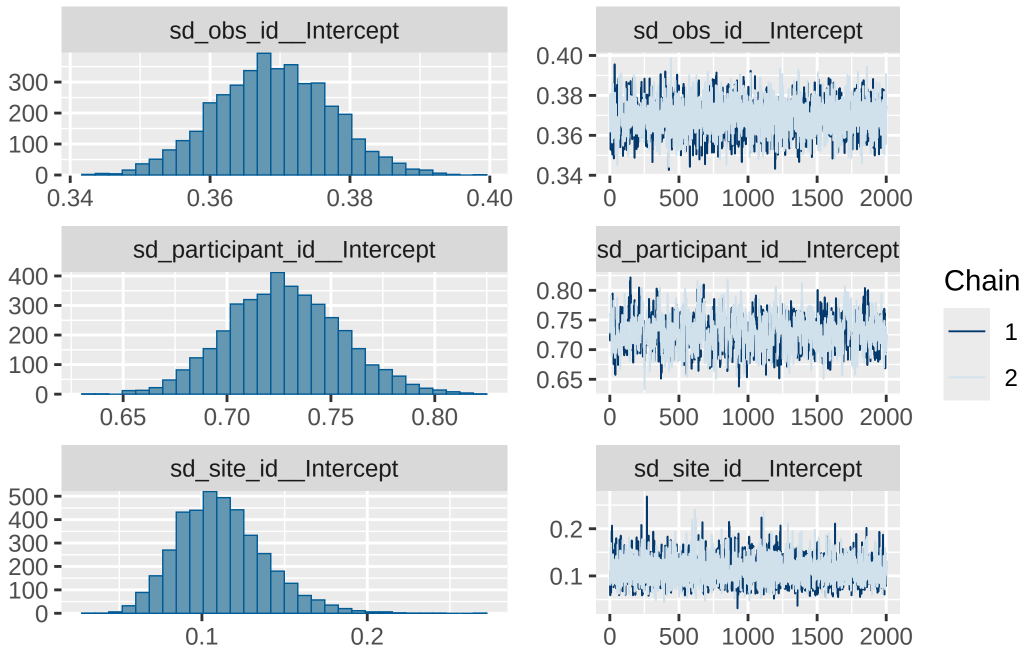
Model 5b effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

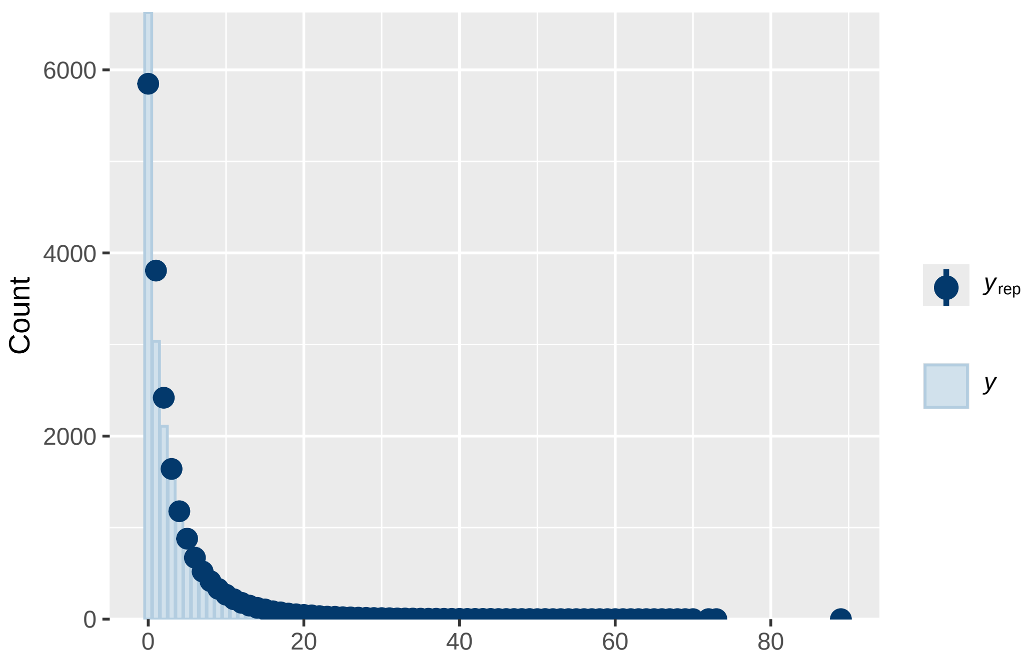
```
# A tibble: 3 x 4
  Variable                                Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>                                      <dbl>          <dbl>          <dbl>
1 Breast development (per 1-unit) [ADHD ~     6.96           4.34           9.69
2 ADHD traits (per 1-unit) [Breast devel~    21.3           20.6           22.1
3 Interaction (Breast development × ADHD~    -0.27          -0.76           0.25
```

Model 5b diagnostics

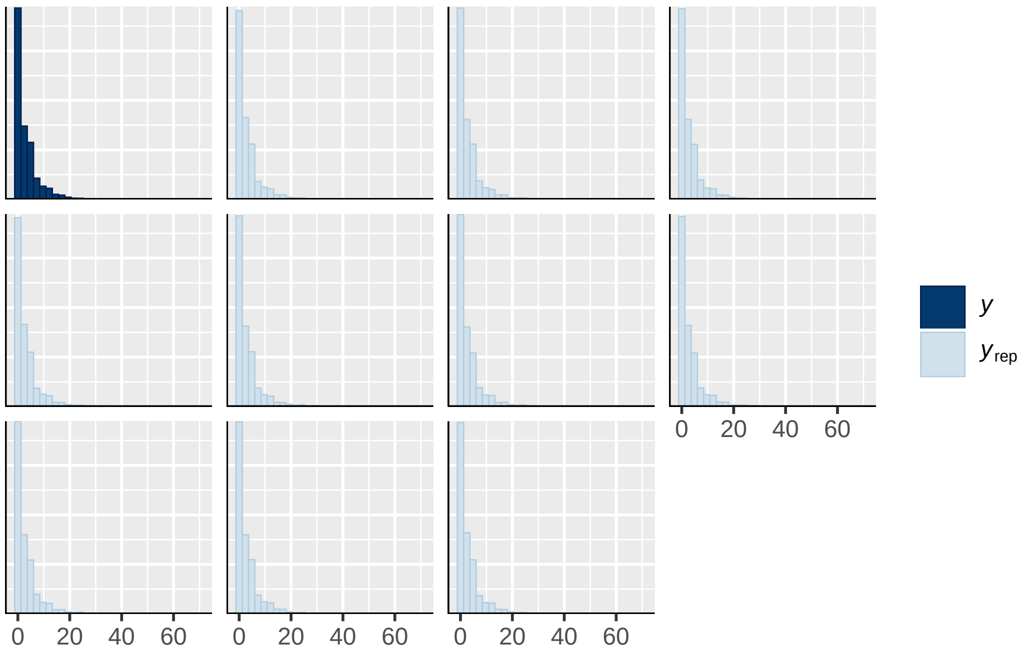
```
                 prior     class                                   coef
          normal(0, 1)         b                                       
          normal(0, 1)         b                         adhd_traits_sc
          normal(0, 1)         b                                 age_sc
          normal(0, 1)         b                breast_development_p_sc
          normal(0, 1)         b breast_development_p_sc:adhd_traits_sc
          normal(0, 1)         b                       ethnraceblack_nh
          normal(0, 1)         b                       ethnracehispanic
          normal(0, 1)         b                          ethnraceother
          normal(0, 1)         b                                 inr_sc
student_t(3, 0.7, 2.5) Intercept                                       
  student_t(3, 0, 2.5)        sd                                       
  student_t(3, 0, 2.5)        sd                                       
  student_t(3, 0, 2.5)        sd                              Intercept
  student_t(3, 0, 2.5)        sd                                       
  student_t(3, 0, 2.5)        sd                              Intercept
  student_t(3, 0, 2.5)        sd                                       
  student_t(3, 0, 2.5)        sd                              Intercept
  student_t(3, 0, 2.5)        sd                                       
  student_t(3, 0, 2.5)        sd                              Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
        obs_id                  0        (vectorized)
        obs_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
0.3 0.4 0.5 0.6
b_Intercept
0 500 1000 1500 2000
b_breast_development_p_sc
0.05 0.10 0.15 0.20
b_adhd_traits_sc
1.04 1.08 1.12 1.16
b_adhd_traits_sc
Chain
1
2
b_age_sc
−0.20 −0.15 −0.10
b_age_sc
b_ethnraceblack_nh
−0.2 −0.1 0.0 0.1
b_ethnraceblack_nh
b_ethnracehispanic
−0.2 −0.1 0.0 0.1
b_ethnracehispanic
b_ethnraceother
−0.2 −0.1 0.0 0.1
b_ethnraceother
b_inr_sc
−0.20 −0.15 −0.10
b_inr_sc
Chain
1
2
−0.10 −0.05 0.00 0.05
sd_family_id__Intercept
0.60 0.65 0.70 0.75 0.80
sd_family_id__Intercept

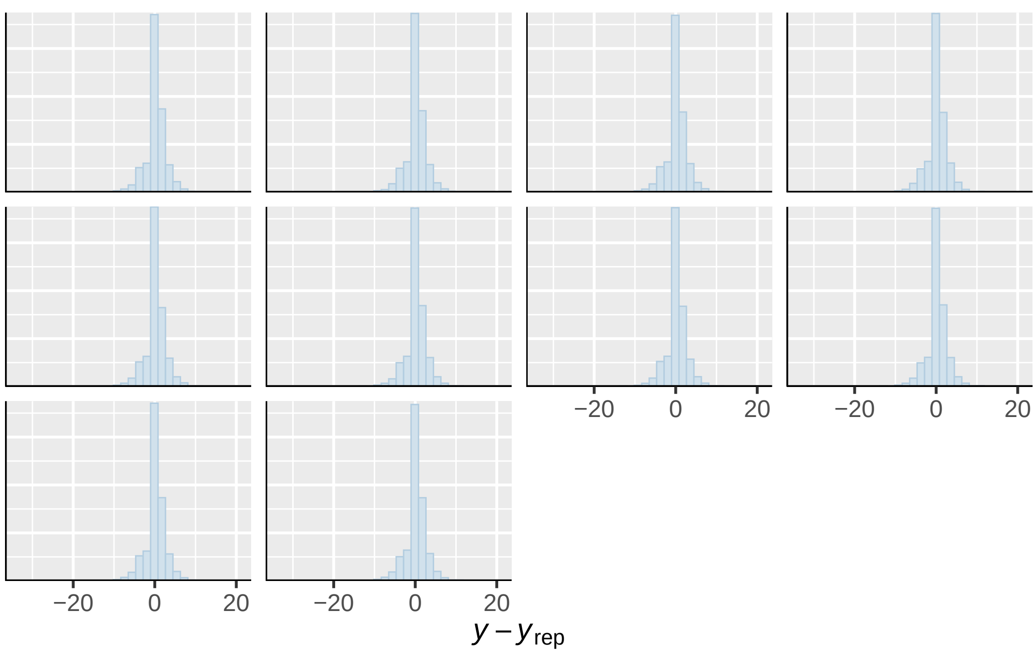




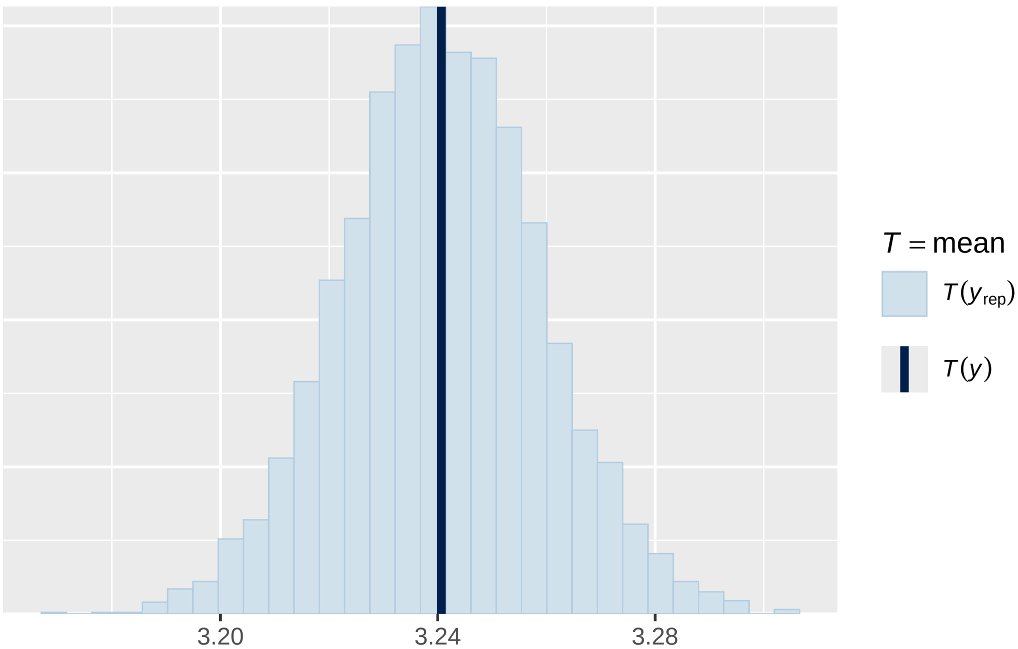
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



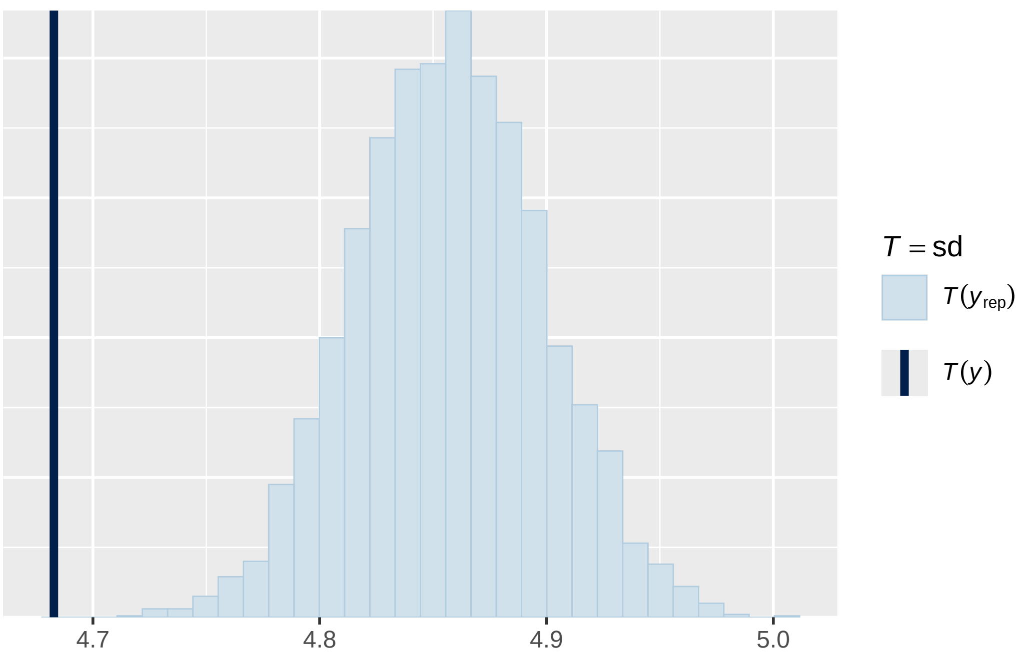
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

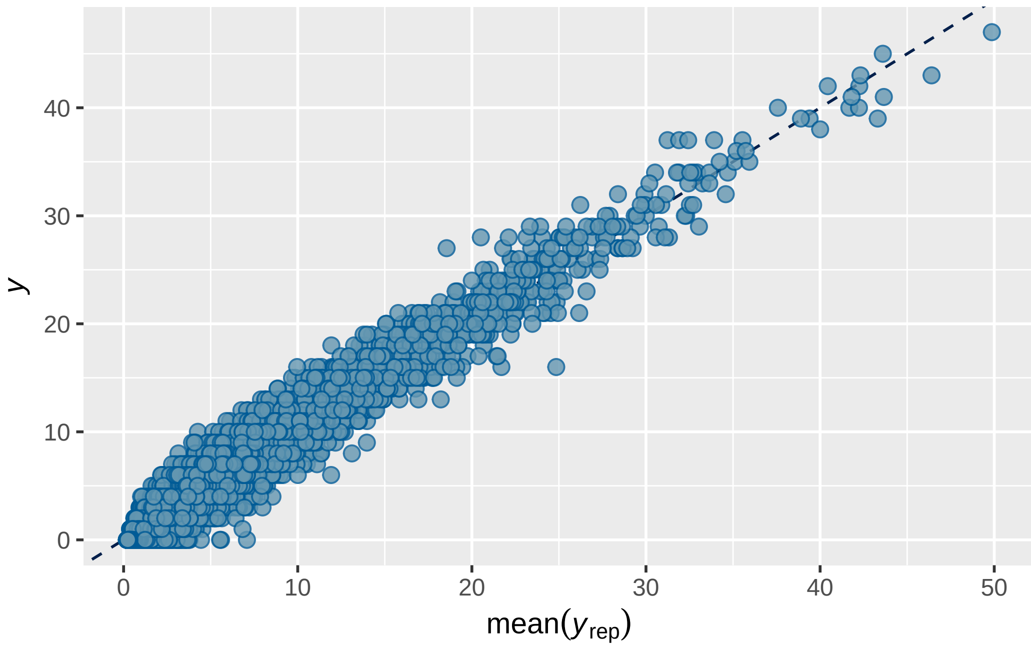


`stat_bin()` using `bins = 30`. Pick better value `binwidth`.




```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




y
mean(y_{rep})
0
10
20
30
40
50